

CS 305: Computer Networks

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Link Layer

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Link layer, LANs: outline

6.1 introduction, services

6.2 error detection, correction

6.3 multiple access protocols

6.4 LANs

- addressing, ARP
- Ethernet
- switches
- VLANs

6.5 link virtualization: MPLS

6.6 data center networking

6.7 a day in the life of a web request

Ethernet

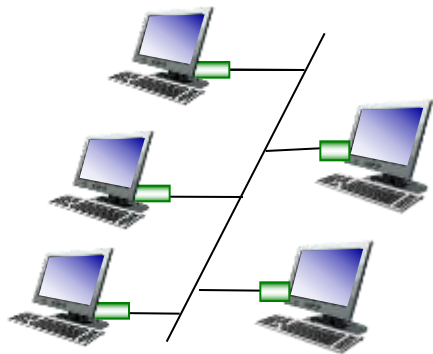
“dominant” wired LAN technology: 简单 + 能规模化

- first widely used LAN technology
- simpler, cheap
- kept up with speed race: 10 Mbps – 10 Gbps

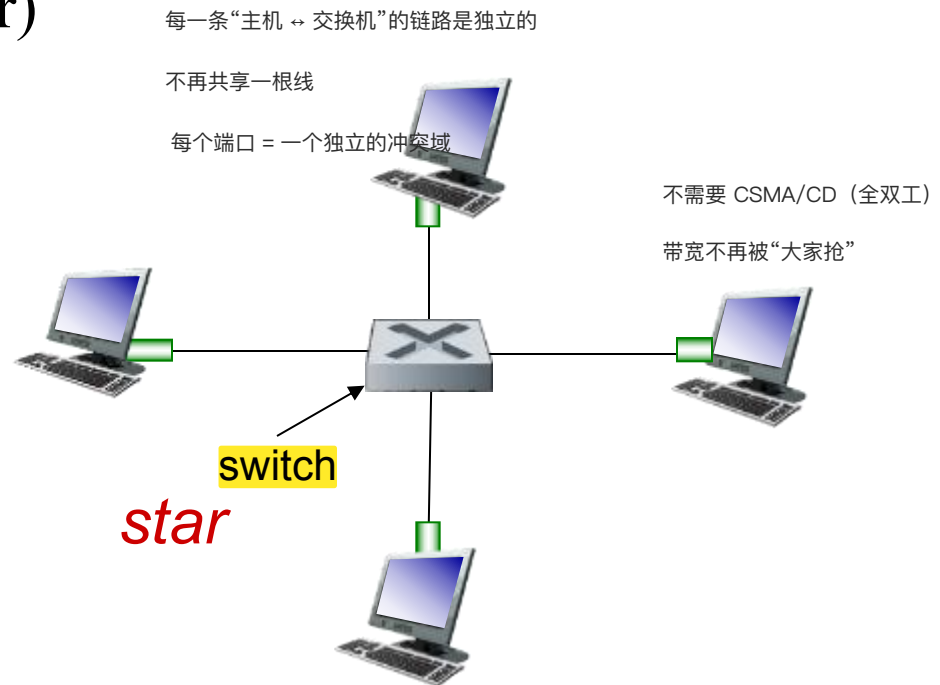
协议核心思想几乎没变，但速度能不断提升

Ethernet: physical topology

- **bus**: popular through mid 90s
 - all nodes in same collision domain (can collide with each other)
- **star**: prevails today
 - active **switch** in center
 - each “spoke” runs a (separate) Ethernet protocol (nodes do not collide with each other)

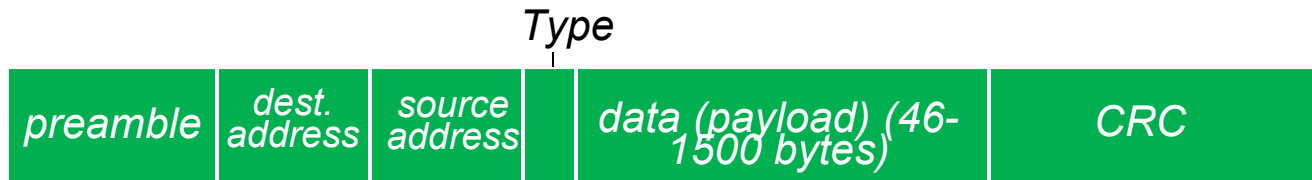


bus: coaxial cable



Ethernet frame structure

sending adapter encapsulates IP datagram (or other network layer protocol packet) in **Ethernet frame**



preamble:

- 7 bytes with pattern 10101010 followed by one byte with pattern 10101011
- Transmit frame at 10Mbps, 100Mbps, 1Gbps
- Drift from target rate
- used to **synchronize** receiver, sender clock rates

让接收方“跟上发送方的时钟节奏”

防止速率偏差导致采样错误

Ethernet frame structure (more)

- *addresses*: 6 byte source, destination MAC addresses
 - if adapter receives frame with matching destination address, or with broadcast address (e.g. ARP packet), it passes data in frame to network layer protocol
 - otherwise, adapter discards frame
- *type*: indicates higher layer protocol (mostly IP but others possible, e.g., Novell IPX, AppleTalk)
- *CRC*: cyclic redundancy check at receiver
 - error detected: frame is dropped

不一致直接丢帧，不重传，不通知；因此ethernet是不可靠的



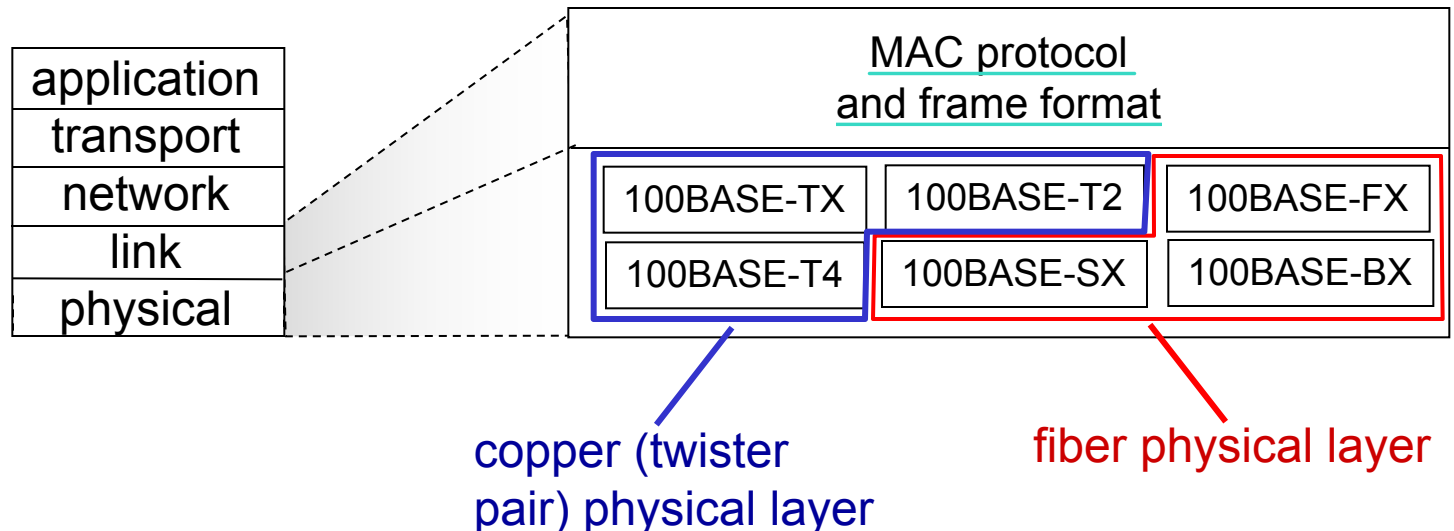
Ethernet: unreliable, connectionless

- *connectionless*: no handshaking between sending and receiving NICs 越靠下层，越不管连接；越靠上层，才谈连接
- *unreliable*: receiving NIC doesn't send acks or nacks to sending NIC 接收方网卡 不会 给发送方回收收到没有，收没收到发送方不知道
 - data in dropped frames recovered **only if initial sender uses higher layer rdt** (e.g., TCP), otherwise dropped data lost 只有上层协议（比如 TCP）负责重传，以太网自己不管
 - **Unaware** of whether it is transmitting a brand-new datagram with brand-new data 以太网只负责把这一帧丢到线上
- Ethernet's MAC protocol: unslotted *CSMA/CD with binary backoff*

802.3 Ethernet standards: link & physical layers

- *many* different Ethernet standards
 - common MAC protocol and frame format
 - different speeds: 2 Mbps, 10 Mbps, 100 Mbps, 1Gbps, 10 Gbps, 40 Gbps
 - different physical layer media: fiber, cable

不管是 10M、100M、1G、10G，以太网“帧格式和 MAC 规则是一样的



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Ethernet switch

■ link-layer device

- store, forward Ethernet frames
- examine incoming frame's MAC address, **selectively** forward frame to one-or-more outgoing links when frame is to be forwarded on segment

交换机收到一个以太网帧：

先把帧收全 (store)

看帧头里的 目的 MAC 地址

决定从哪个端口发出去 (forward)

交换机只转发到“需要的那条线”，不会像 hub 一样全网广播（除非它不知道目的 MAC 在哪）

■ *transparent*

- hosts are unaware of presence of switches

电脑（主机）根本不需要知道交换机存在

■ *plug-and-play, self-learning*

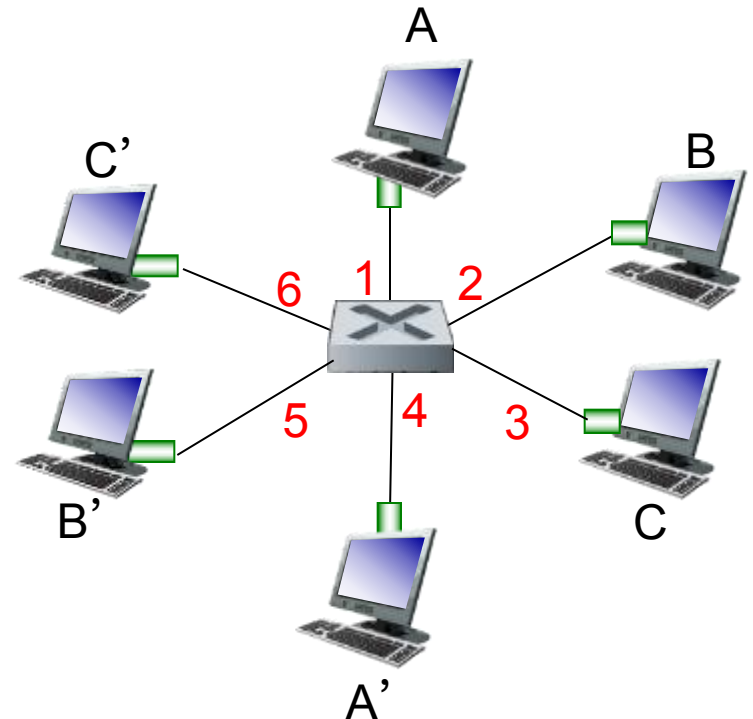
- switches do not need to be configured

看收到的帧的源 MAC”，就知道“这个 MAC 在这个端口后面

Switch: *multiple* simultaneous transmissions

每台主机到交换机是独享链路

- hosts have **dedicated, direct** connection to switch
- switches **buffer** packets
- Ethernet protocol used on *each* incoming link, but no collisions; **full duplex**
- *switching*: A-to-A' and B-to-B' can transmit simultaneously, without collisions



switch with six interfaces
(1,2,3,4,5,6)

Switch forwarding table

交换机从端口 i 收到一个帧，它看帧头的 源 MAC = X
记一条：X 在端口 i

Q: how does switch know A
reachable via interface 4, B'
reachable via interface 5?

如果表里没有 Y → 泛洪 flood (除来口外所有口都发一遍), 直到有人回应, 交换机就学到了

A: each switch has a **switch table**, each entry:

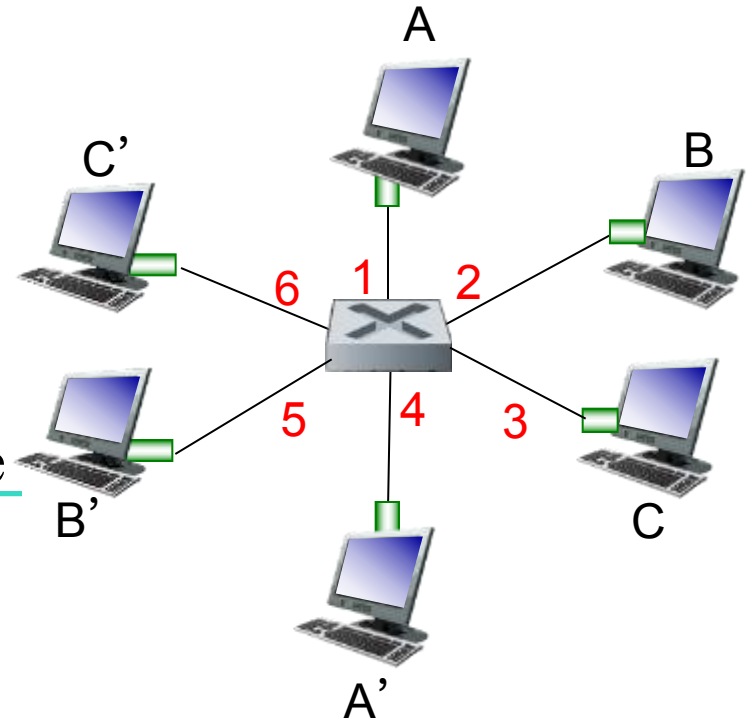
- (MAC address of host, interface to reach host, time stamp)
- looks like a routing table!

Q: how are entries created, maintained in switch table?

- something like a routing protocol?

不是路由协议! 交换机不跑 RIP/OSPF

就是通过学习

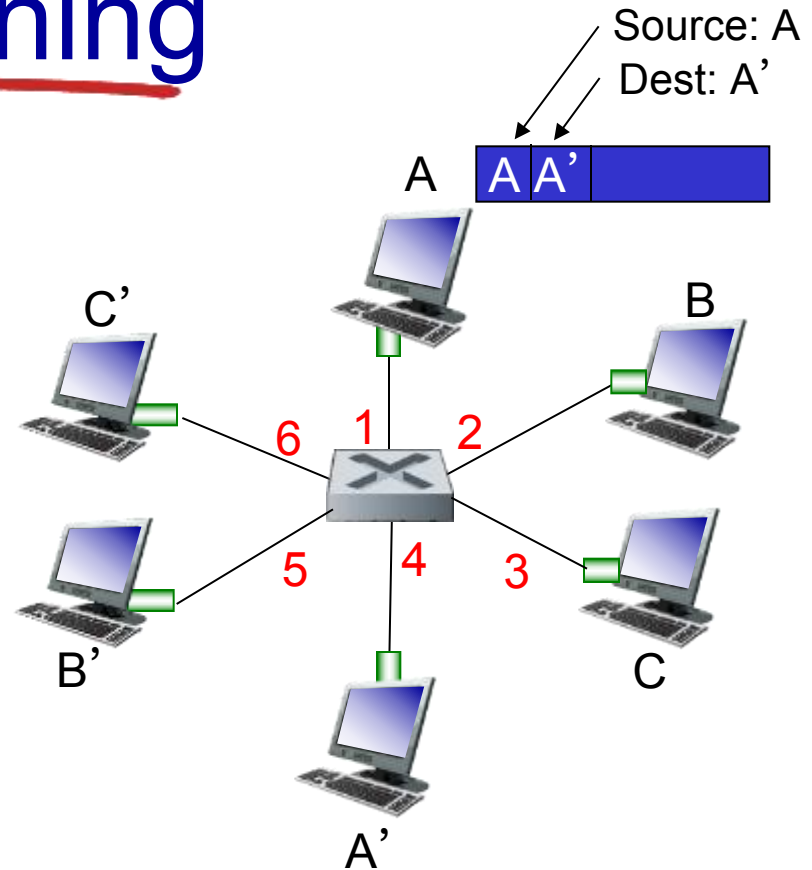


switch with six interfaces
(1,2,3,4,5,6)

Switch: self-learning

Switch *learns* which hosts can be reached through which interfaces

- when frame received, switch “learns” location of sender: incoming LAN segment
- records sender/location pair in switch table

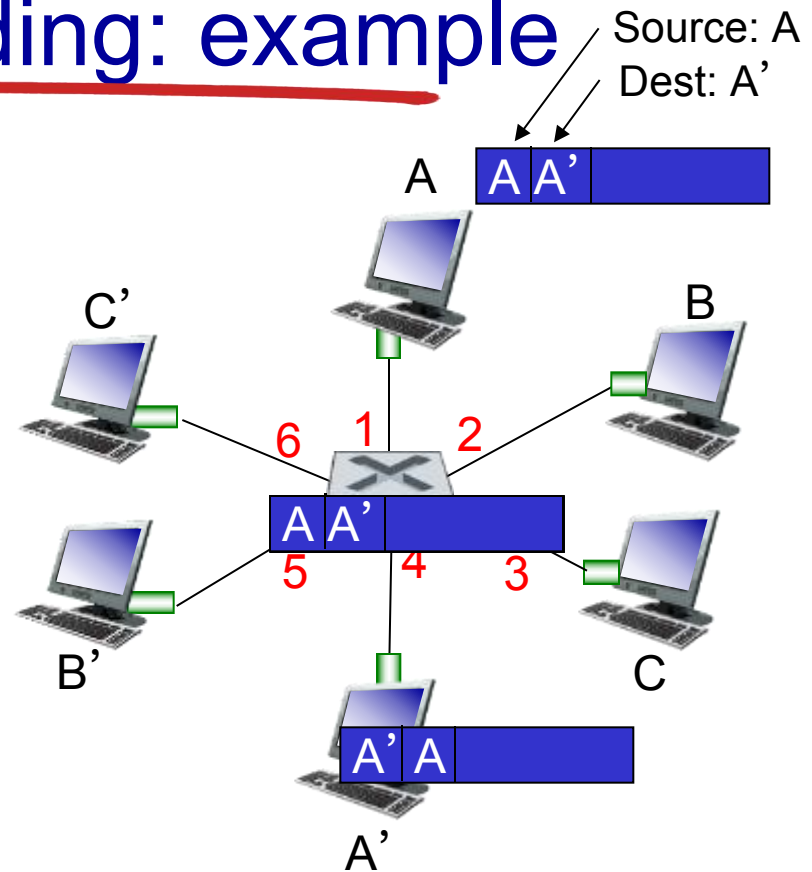


MAC addr	interface	TTL
A	1	60

*Switch table
(initially empty)*

Self-learning, forwarding: example

- frame destination, A',
location unknown: *flood*
- destination A location
known: *selectively send*
on just one link



MAC addr	interface	TTL
A	1	60
<u>A'</u>	<u>4</u>	<u>60</u>

*switch table
(initially empty)*

Self-learning, forwarding: example

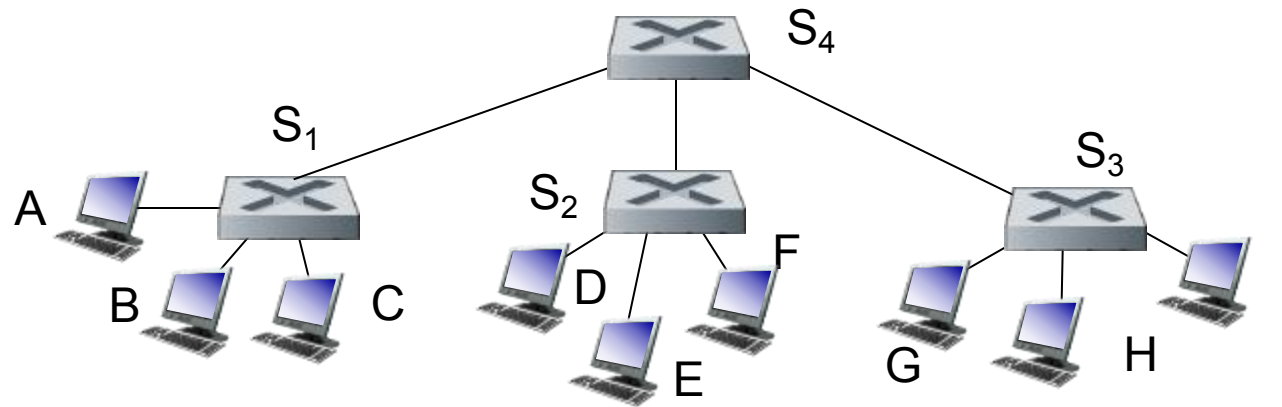
- 1 收帧先学源 MAC (源 MAC → 入端口)
- 2 查目的 MAC
- 3 不知道 → flood (除来口)
- 4 知道 → 只走一个口; 同口 → 直接丢

Suppose a frame with destination address DDDD-DD-DD-DD-DD arrives at the switch **on interface x**.

- **no entry** in the table for DD-DD-DD-DD-DD-DD:
 - forwards copies of the frame to the output buffers preceding all interfaces except for interface x.
- an entry in the table associating DD-DD-DD-DD-DD-DD **with interface x**:
 - 目的 MAC 对应的接口 = 来的接口, 不用再转发, 直接丢弃frame
 - the switch **performs the filtering function by discarding the frame**.
- an entry in the table associating DD-DD-DD-DD-DD-DD with **interface y**:
 - frame needs to be forwarded to the LAN segment attached to interface y.

Interconnecting switches

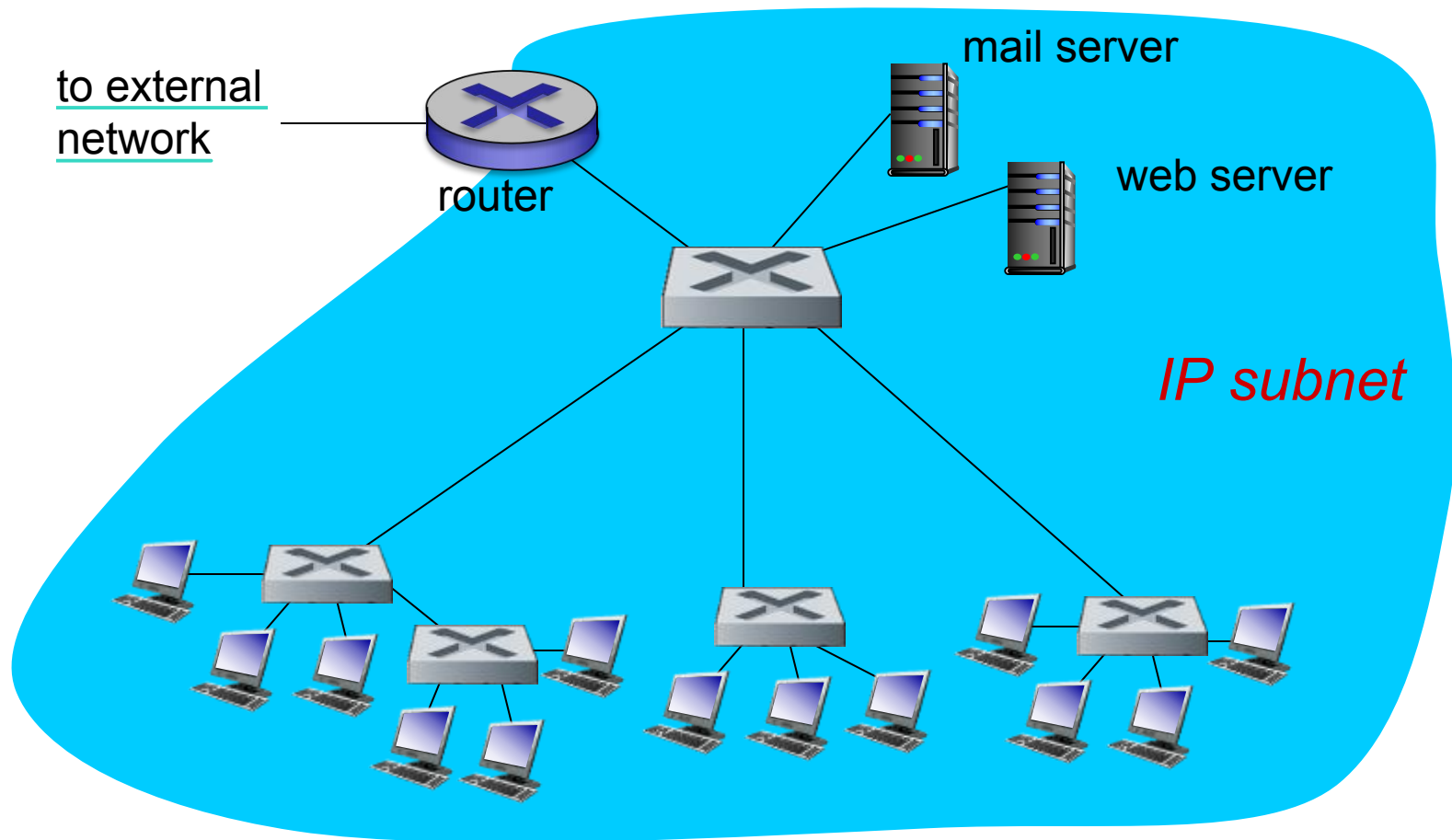
self-learning switches can be connected together:



Q: sending from A to G - how does S₁ know to forward frame destined to G via S₄ and S₃?

- A: self learning! (works exactly the same as in single-switch case!)

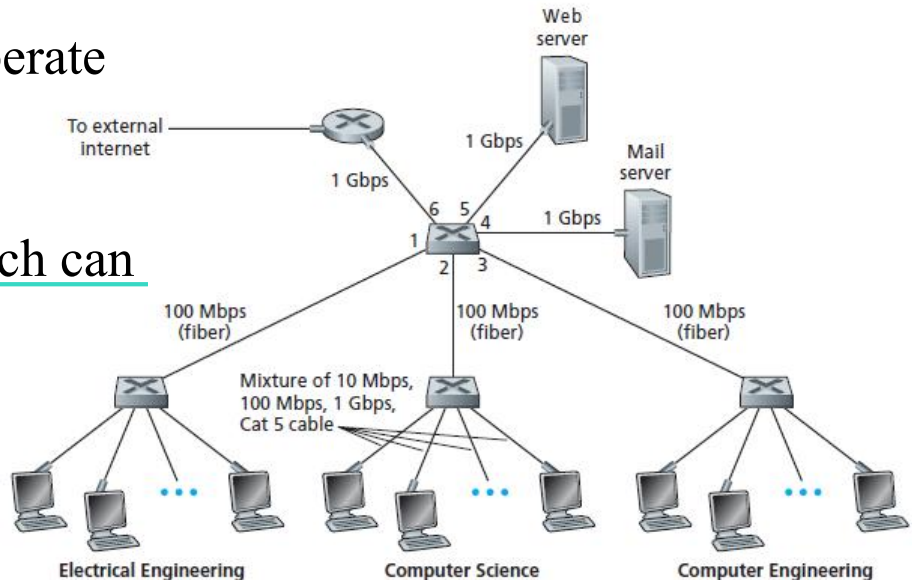
Institutional network



Properties of link-layer Switching

When compared with bus or hub:

- **Elimination of collisions**
 - Buffer frames; never transmit more than one frame on a segment at a time
- **Heterogeneous links**
 - Switch can isolates one link from another
 - Different links in the LAN can operate at different speeds and media
- **Management**
 - If one NIC malfunctions, the switch can detect it and disconnect it.



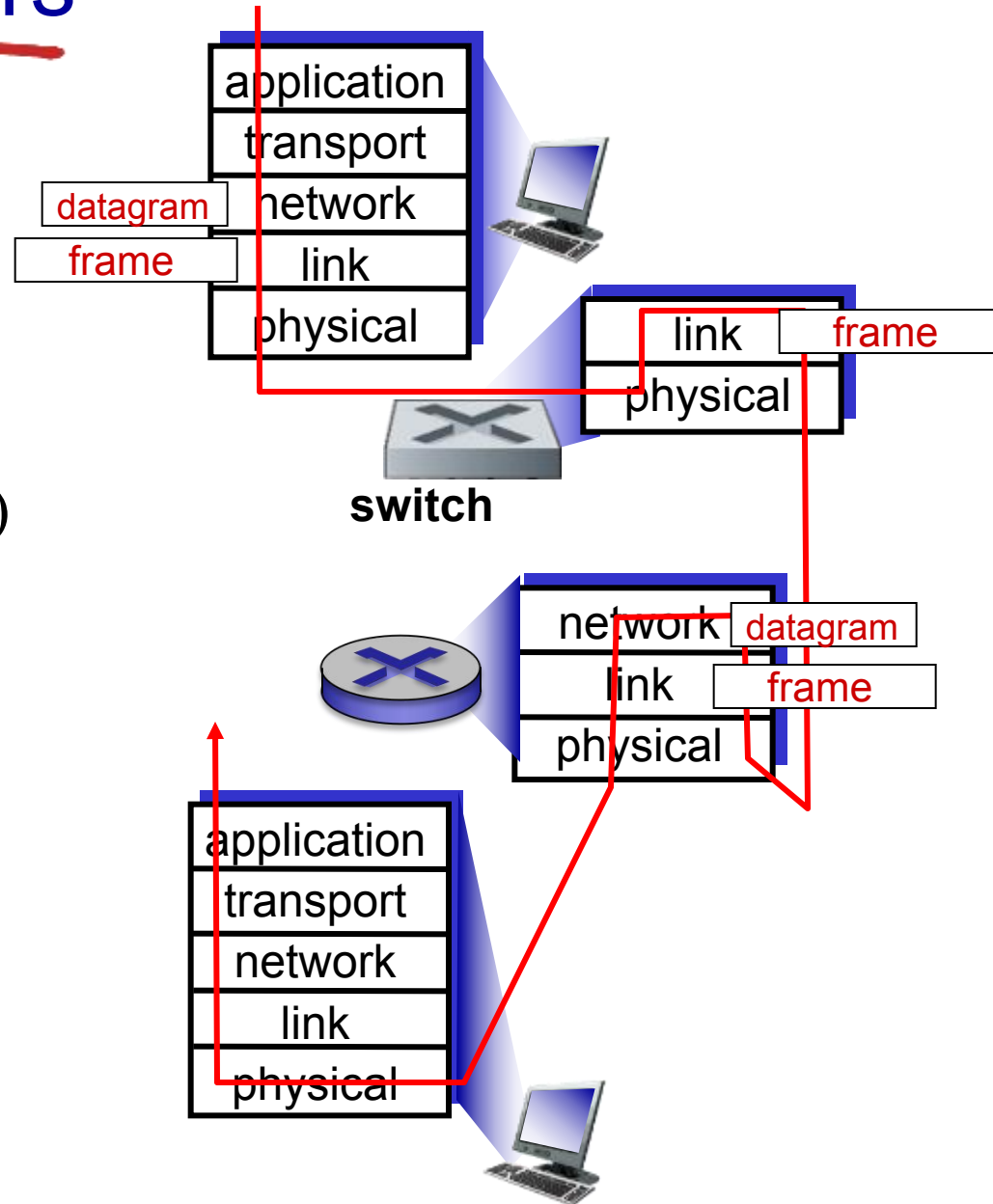
Switches vs. routers

both are **store-and-forward**:

- **routers**: network-layer devices (examine network-layer headers)
- **switches**: link-layer devices (examine link-layer headers)

both have forwarding tables:

- **routers**: **compute tables using routing algorithms**, IP addresses
- **switches**: learn forwarding table using flooding, learning, MAC addresses



Switches vs. routers

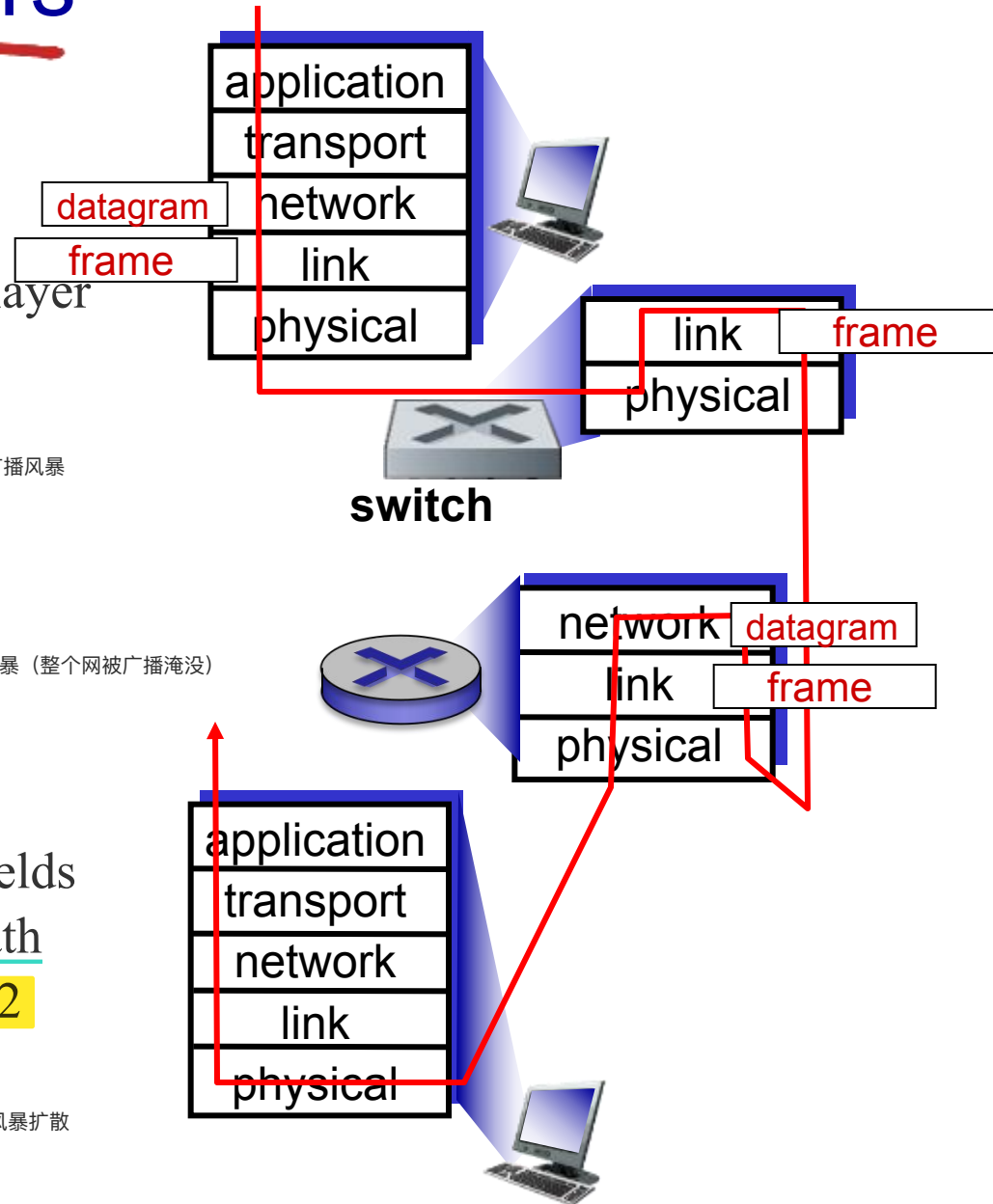
Switches:

- **plug-and-play**
- Process frames only up through layer 2, relatively high filtering and forwarding rates
- **a spanning tree** 因为二层网络如果交换机环路会产生广播风暴
- a large switched network would require large ARP
- broadcast storms 二层广播太多/环路会导致广播风暴（整个网被广播淹没）

Routers

- not plug-and-play
- process up through the layer-3 fields
- rich topology; choose the best path
- **firewall protection against layer-2 broadcast storms.**

路由器天然把广播域隔开（不同子网不互相转发二层广播），能“挡住”二层广播风暴扩散



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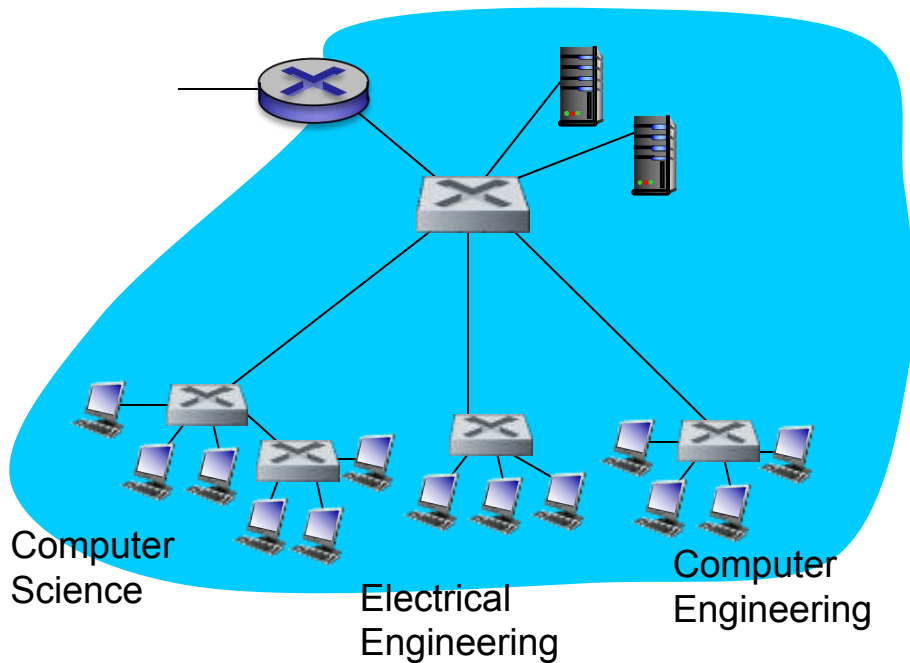
6.4 LANs

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- Ethernet
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VLANs: motivation



consider:

- CS user moves office to EE, but wants to connect to CS switch
- Inefficient use of switches
- **single broadcast domain:**
 - all layer-2 broadcast traffic (ARP, DHCP, unknown location of destination MAC address) must cross entire LAN
 - security/privacy, efficiency issues

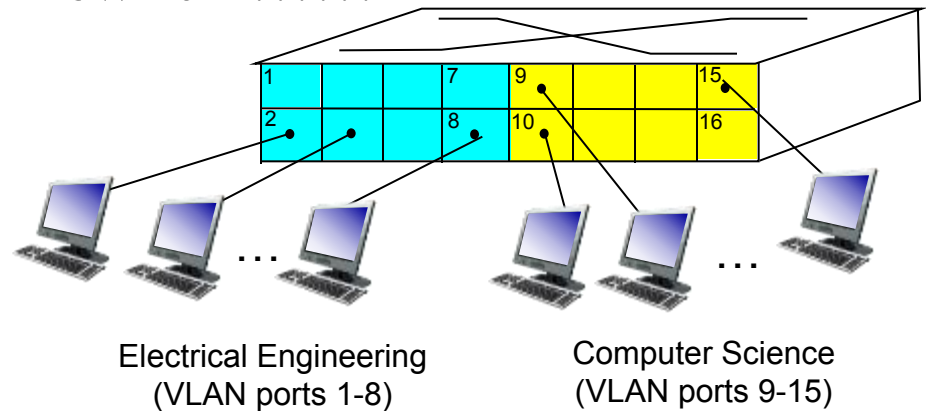
VLANs

Virtual Local Area Network

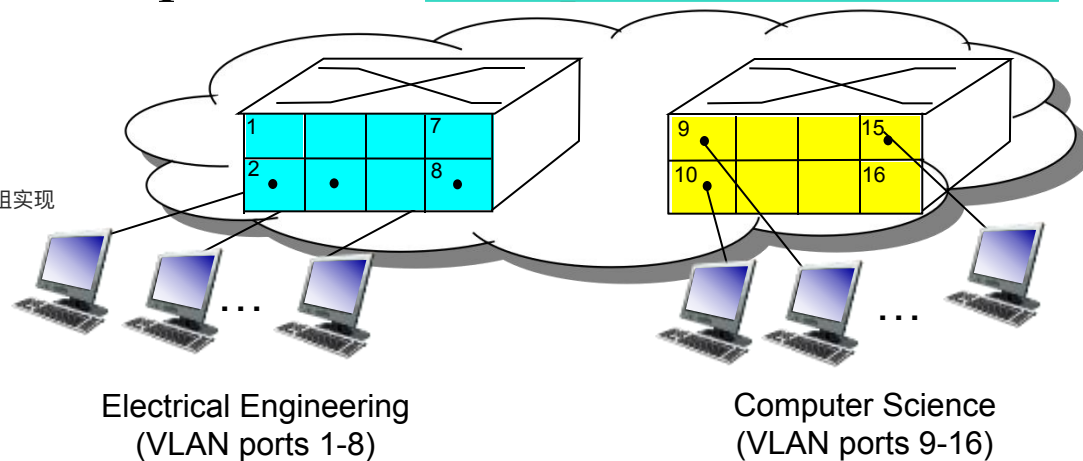
define multiple **virtual** LANS over single physical LAN infrastructure.

VLAN 的本质就是“二层隔离 + 广播域隔离”，靠管理员把端口划到不同组实现

port-based VLAN: switch ports grouped (by switch management software) so that *single* physical switch

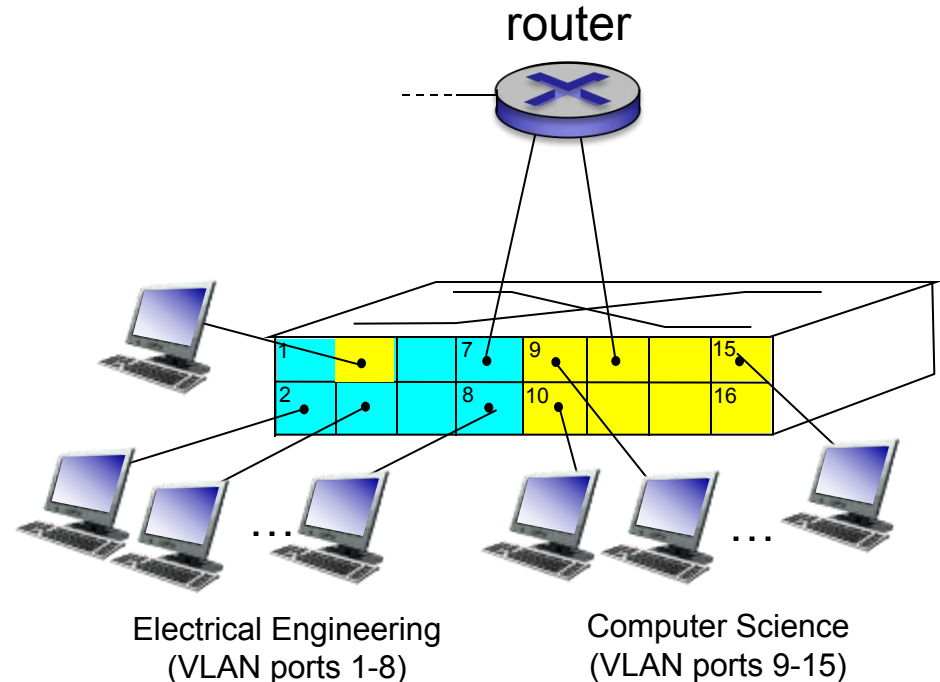


... operates as **multiple** virtual switches



Port-based VLAN

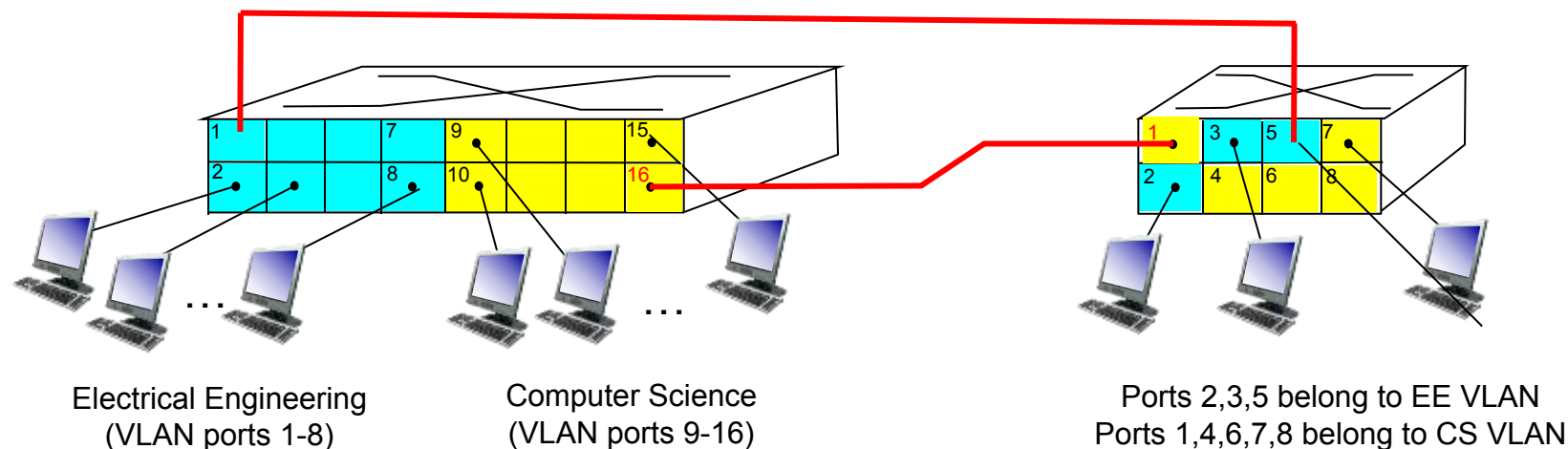
- **traffic isolation**: frames to/from ports 1-8 can *only* reach ports 1-8
 - can also define VLAN based on MAC addresses of endpoints, rather than switch port
- **dynamic membership**: ports can be dynamically assigned among VLANs
- **forwarding between VLANs**: done via **routing** (just as with separate switches)
 - in practice vendors sell combined switches plus routers



不同 VLAN 之间，二层默认不互通

如果 EE VLAN 要访问 CS VLAN，必须“上升到三层”：通过**路由（routing）**实现
（就像两台独立交换机之间通信一定要过路由器一样）

VLANs spanning multiple switches



- *If some of the CS and EE faculties are in another building, how to connect two switches together as two VLANs?*
 - Two links connect both CS VLAN and EE VLAN.

第一页的做法（不理想的直观方案）

“Two links connect both CS VLAN and EE VLAN.”

意思是：你可以为 EE VLAN 单独拉一条交换机间链路，再为 CS VLAN 单独拉一条链路。

缺点：每个 VLAN 都要占一条物理链路，VLAN 多了就炸了（线、端口都不够）。

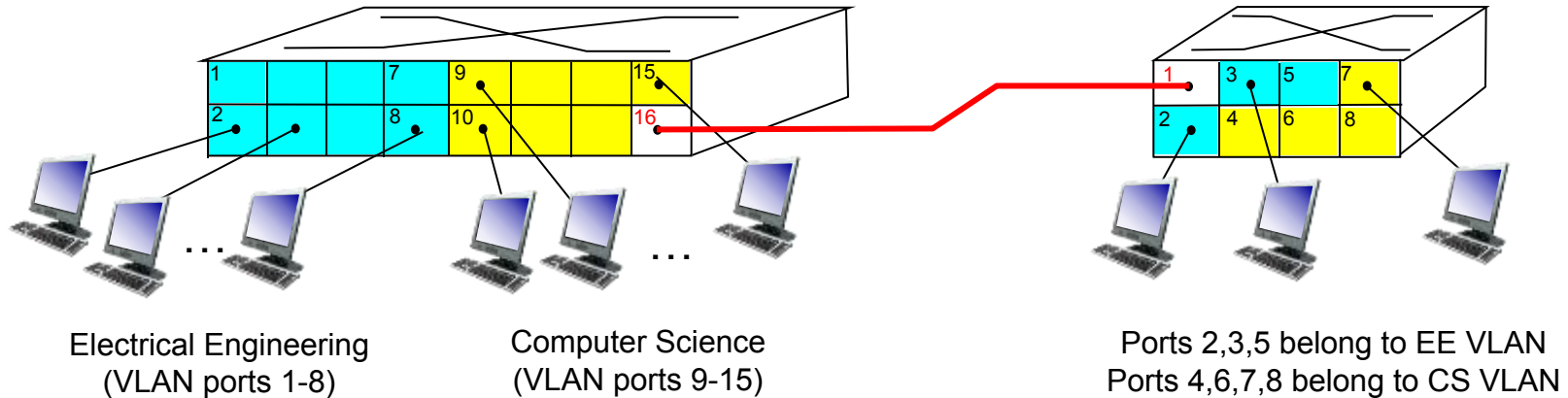
第二页引出 trunk port（更标准的做法）

trunk port（干道口）：交换机之间用一条链路承载多个 VLAN 的流量。

关键问题：同一条链路上传来一帧，以太网原始帧（“vanilla 802.1 frame”）本身不带 VLAN 信息，对端交换机怎么知道它属于哪个 VLAN？

答：必须把 VLAN ID 带上。

VLANs spanning multiple switches



- **trunk port:** carries frames between VLANs defined over multiple physical switches

交换机之间用一条链路承载多个 VLAN 的流量

- frames forwarded within VLAN between switches can't be vanilla 802.1 frames (must carry **VLAN ID info**)
- Extended Ethernet frame format: **802.1q** protocol adds/removed additional header fields for frames forwarded between trunk ports

同一条链路上传来一帧，以太网原始帧（“vanilla 802.1 frame”）本身不带 VLAN 信息，对端交换机怎么知道它属于哪个 VLAN

802.1Q 通过在以太网帧里**增加一个额外的标签字段（tag）**来实现

在 trunk 口：发送时加 tag，接收端读 tag 识别 VLAN

在接入终端的 access 口：通常对终端发/收的是“不带 tag 的普通帧”，交换机内部用 VLAN 配置来归

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Data center networks

- 10's to 100's of thousands of hosts, often closely coupled, in close proximity:
 - e-business (e.g. Amazon)
 - content-servers (e.g., YouTube, Akamai, Apple, Microsoft)
 - search engines, data mining (e.g., Google)
- challenges:
 - multiple applications, each serving massive numbers of clients
 - managing/balancing load, avoiding processing, networking, data bottlenecks

服务器之间频繁通信，而不只是“客户端 ↔ 服务器”



Inside a 40-ft Microsoft container,
Chicago data center

Data center networks

load balancer: application-layer routing

- receives external client requests
- directs workload within data center
- returns results to external client (hiding data center internals from client)

把结果返回给客户端 (隐藏内部结构)

对外:

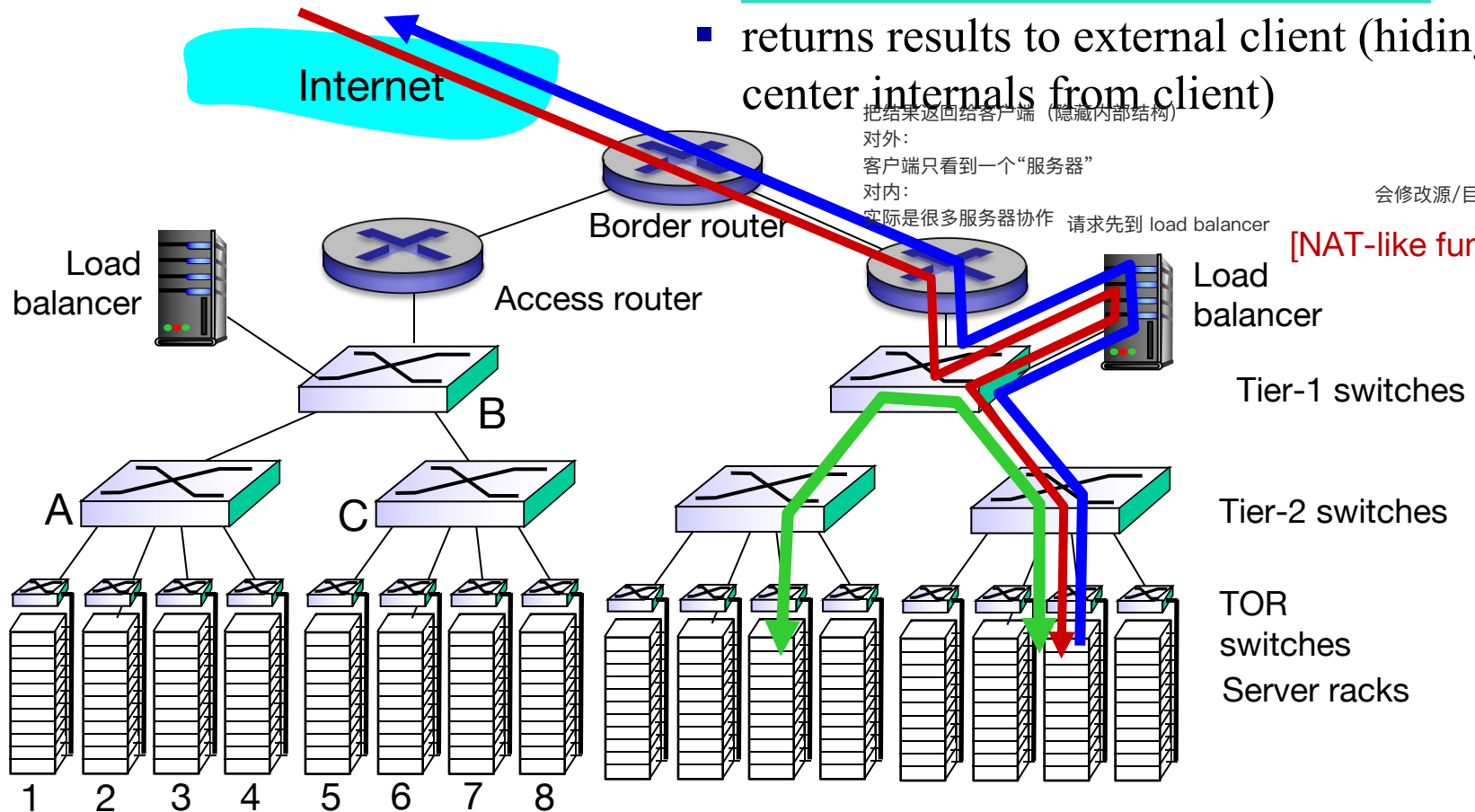
客户端只看到一个“服务器”

对内:

实际是很多服务器协作 请求先到 load balancer

会修改源/目的地址

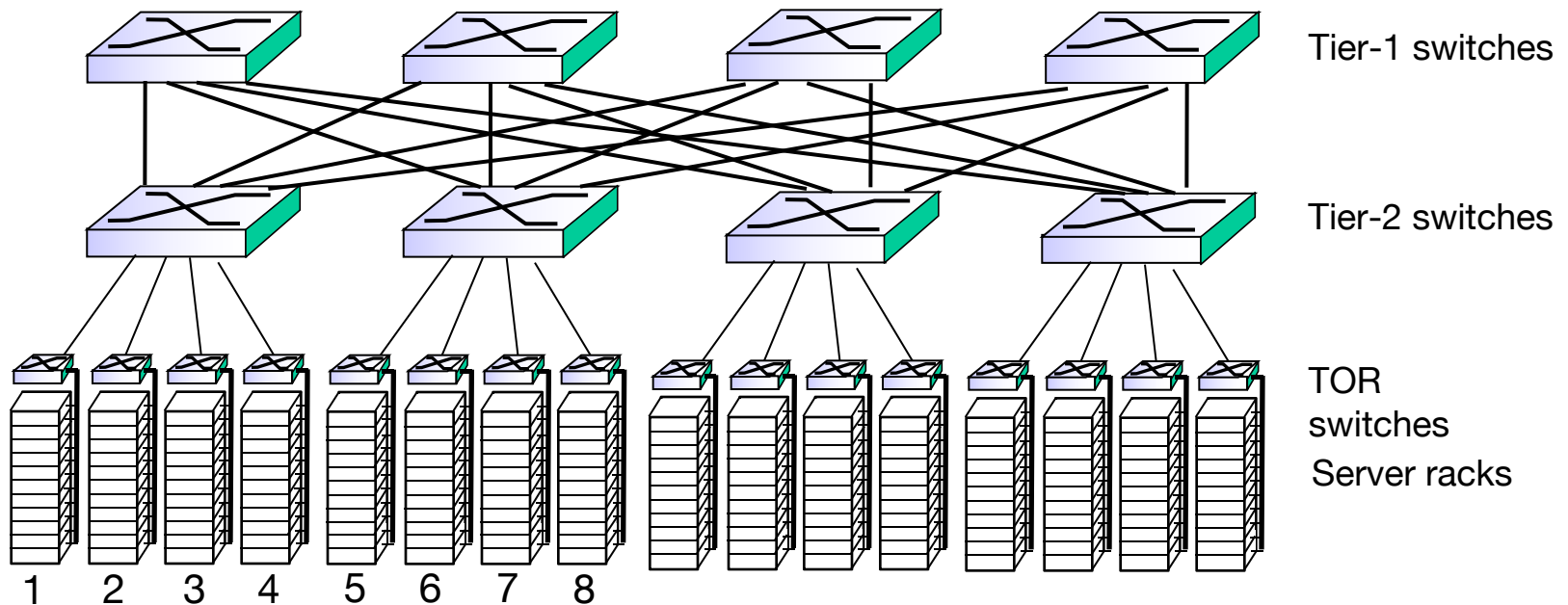
[NAT-like function]



(Hosts are stacked in racks)

Data center networks

- rich interconnection among switches, racks:
 - **increased throughput between racks** (multiple routing paths possible)
 - increased reliability via **redundancy**



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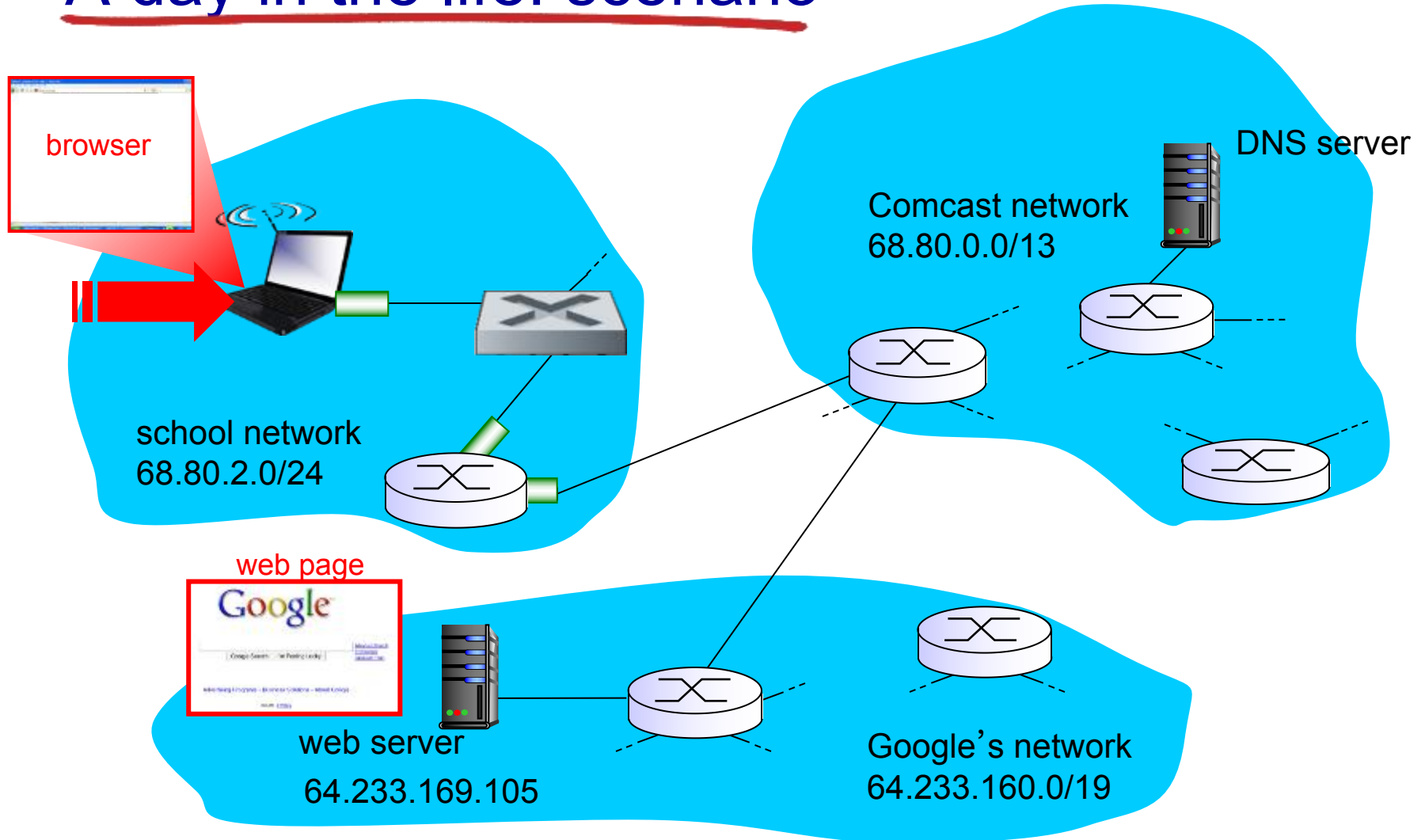
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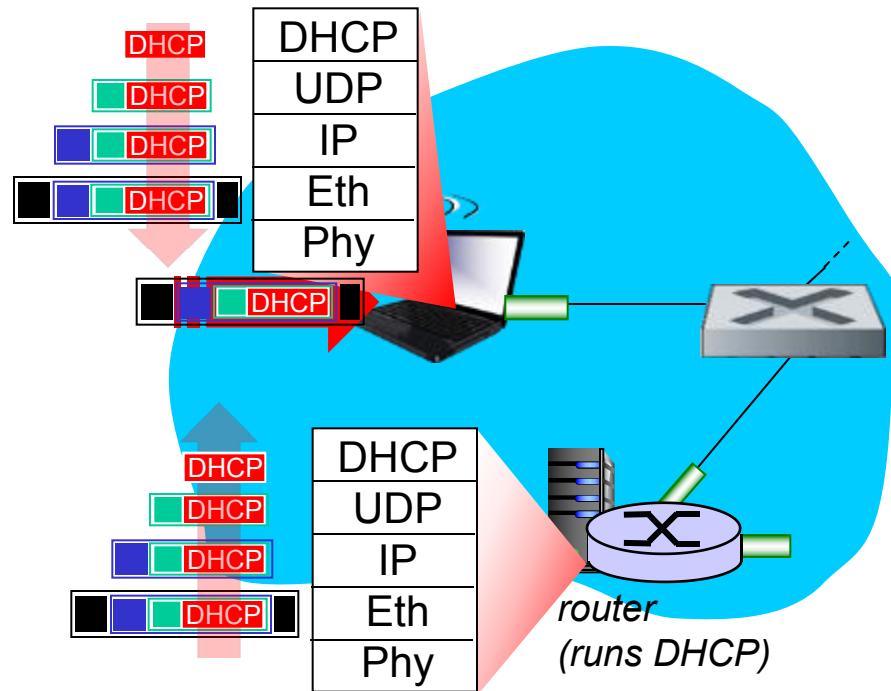
Synthesis: a day in the life of a web request

- journey down protocol stack complete!
 - application, transport, network, link
- putting-it-all-together: synthesis!
 - *goal*: identify, review, understand protocols (at all layers) involved in seemingly simple scenario: requesting www page
 - *scenario*: student attaches laptop to campus network, requests/receives www.google.com

A day in the life: scenario



A day in the life... connecting to the Internet

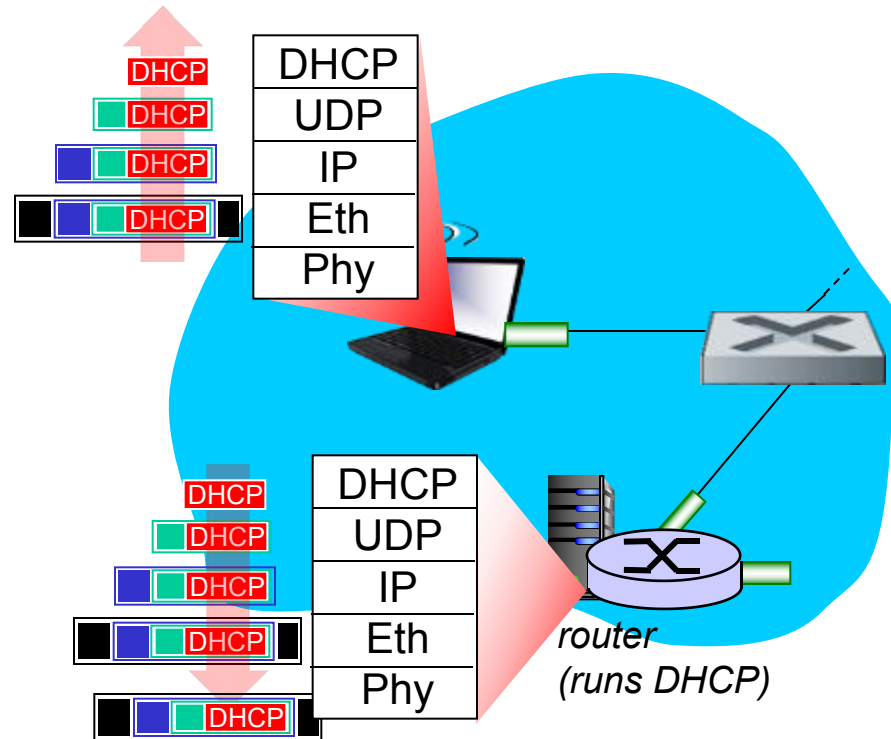


- connecting laptop needs to get its own IP address, addr of first-hop router, addr of DNS server: use *DHCP*

应用层协议

- DHCP request encapsulated in **UDP**, encapsulated in **IP**, encapsulated in **802.3 Ethernet**
- Ethernet frame **broadcast** (dest: FFFFFFFFFFFFFFFF) on LAN, received at router running **DHCP** server
- Ethernet **demuxed** to IP demuxed, UDP demuxed to DHCP

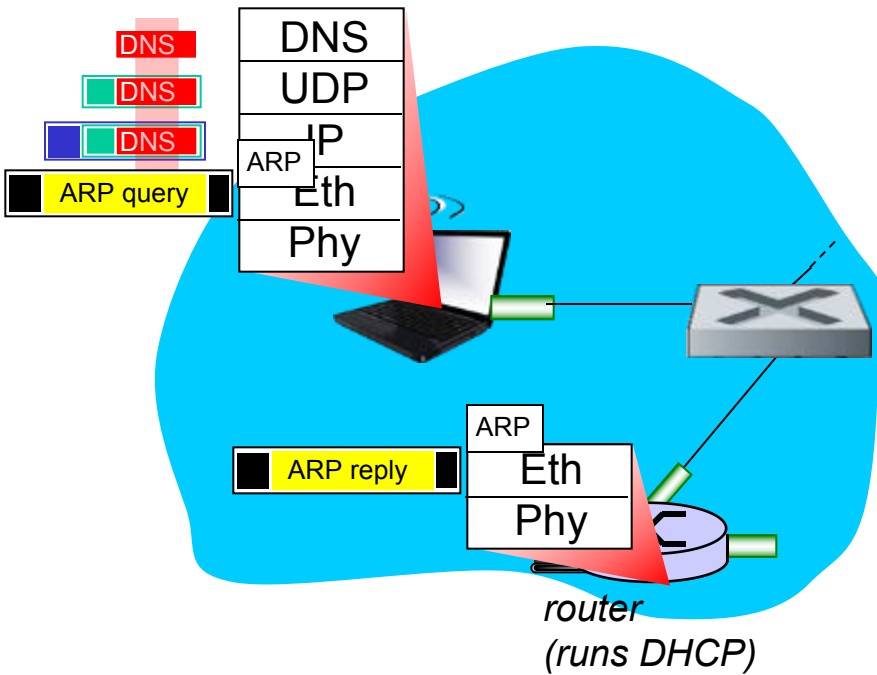
A day in the life... connecting to the Internet



- DHCP server formulates **DHCP ACK** containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulation at DHCP server, frame forwarded (**switch learning**) through LAN, demultiplexing at client
- DHCP client receives DHCP ACK reply

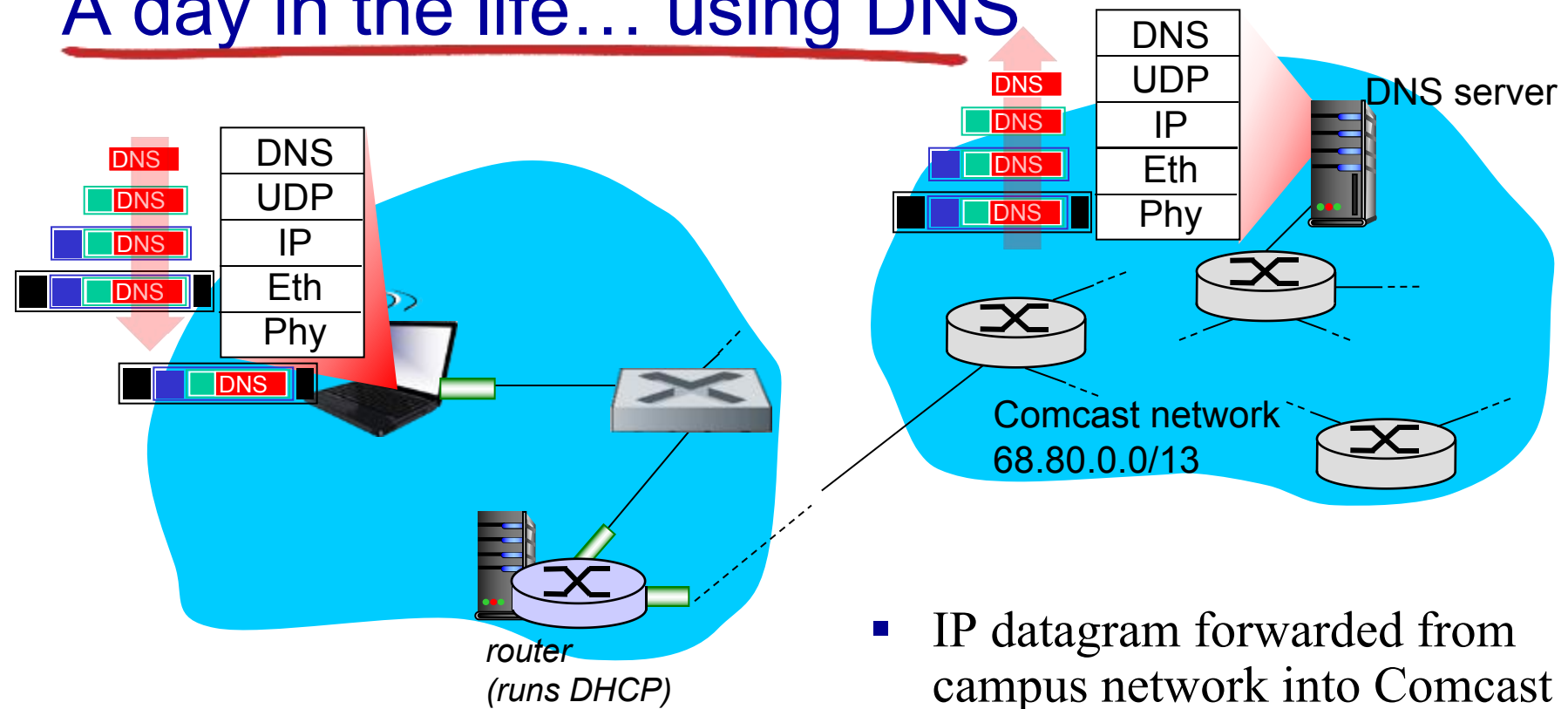
Client now has IP address, knows name & addr of DNS server, IP address of its first-hop router

A day in the life... ARP (before DNS, before HTTP)



- before sending *HTTP* request, need IP address of `www.google.com`:
DNS
- DNS query created, encapsulated in UDP, encapsulated in IP, encapsulated in Eth. To send frame to router, need MAC address of router interface: ARP
- ARP query broadcast, received by router, which replies with **ARP reply** giving MAC address of router interface
- client now knows MAC address of first hop router, so can now send frame containing DNS query

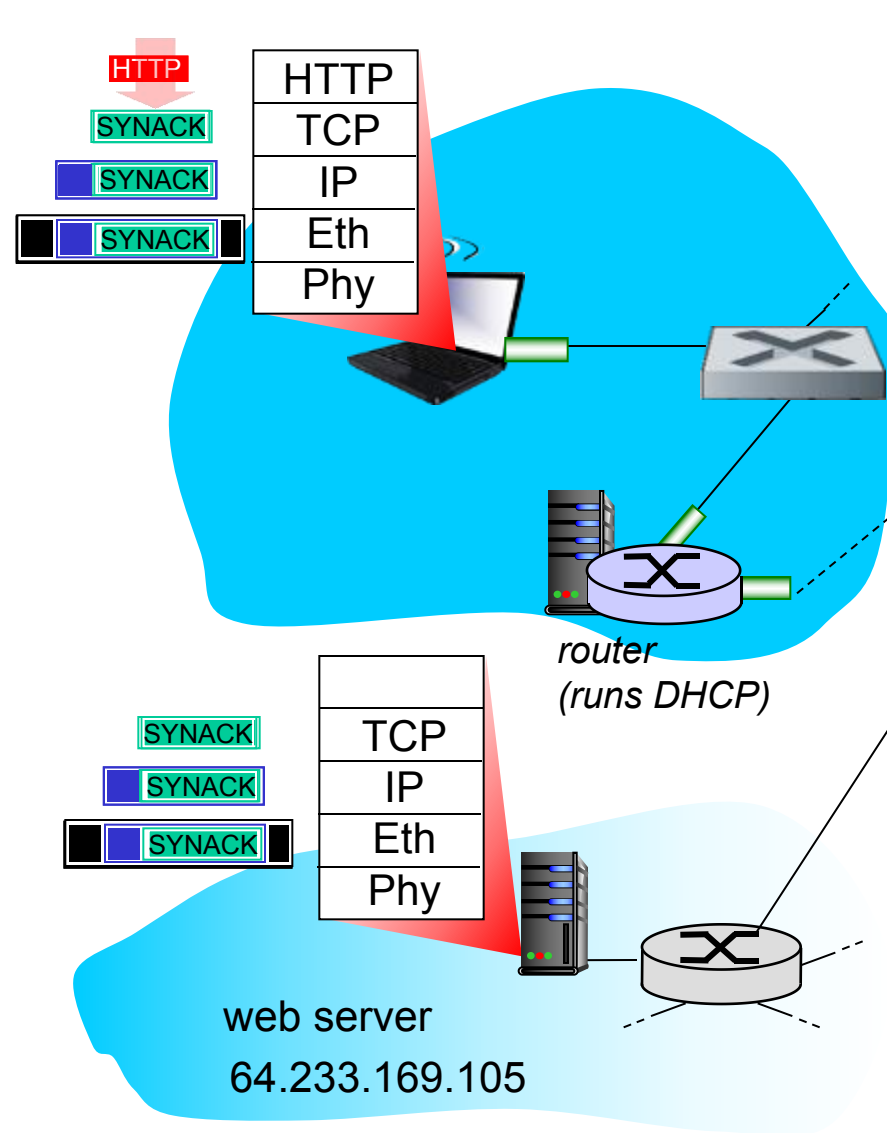
A day in the life... using DNS



- IP datagram containing DNS query forwarded via LAN switch from client to 1st hop router

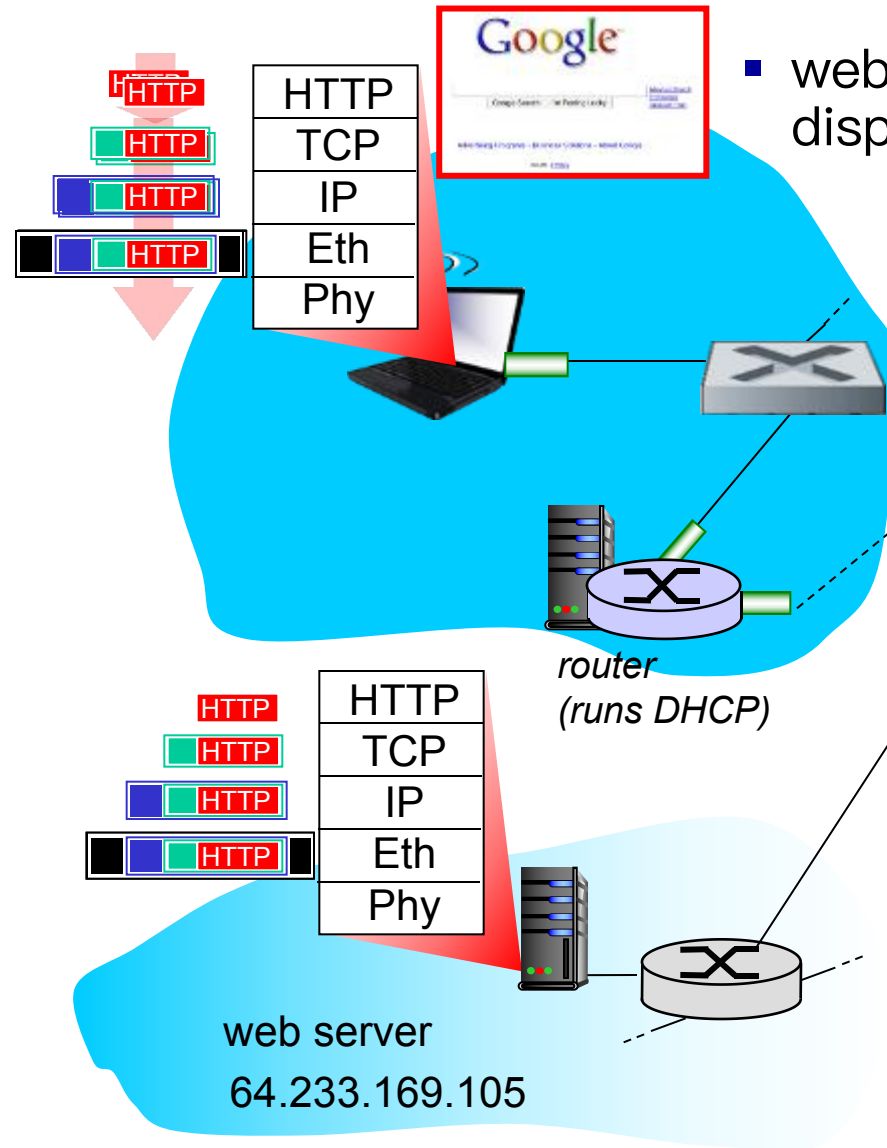
- IP datagram forwarded from campus network into Comcast network, routed (tables created by **RIP**, **OSPF**, **IS-IS** and/or **BGP** routing protocols) to DNS server
- demuxed to DNS server
- DNS server replies to client with IP address of www.google.com

A day in the life...TCP connection carrying HTTP



- to send HTTP request, client first opens **TCP socket** to web server
- TCP SYN segment (step 1 in 3-way handshake) inter-domain routed to web server
- web server responds with TCP SYNACK (step 2 in 3-way handshake)
- TCP connection established!

A day in the life... HTTP request/reply



- web page **finally (!!!)** displayed

- HTTP request sent into TCP socket
- IP datagram containing HTTP request routed to www.google.com
- web server responds with HTTP reply (containing web page)
- IP datagram containing HTTP reply routed back to client

Ch. 7: Wireless and Mobile Networks

Background:

- Number of **wireless (mobile) phone subscribers** now exceeds number of wired phone subscribers (5-to-1)!
- Number of wireless Internet-connected devices equals number of wireline Internet-connected devices
 - laptops, Internet-enabled phones promise anytime untethered Internet access
- two important (but different) challenges
 - *wireless*: communication over wireless link
 - *mobility*: handling the mobile user who changes point of attachment to network

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 - *wireless*: communication over wireless link
 - *mobility*: handling the mobile user who changes point of attachment to network

Chapter 7 outline

7.1 Introduction

Wireless

7.2 Wireless links, characteristics

- CDMA

7.3 IEEE 802.11 wireless LANs (“Wi-Fi”)

7.4 Cellular Internet Access

- architecture
- standards (e.g., 3G, LTE)

Mobility

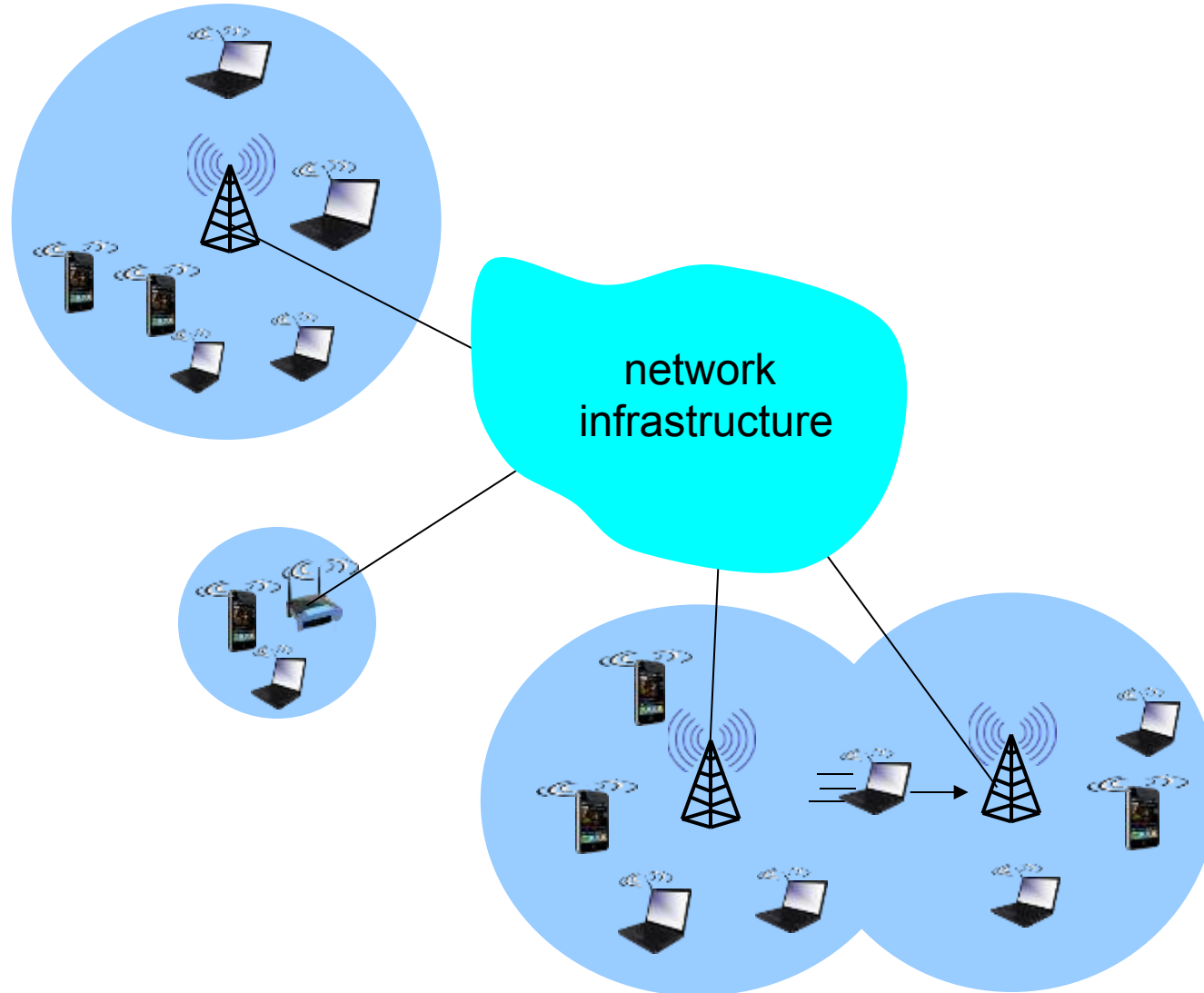
7.5 Principles: addressing and routing to mobile users

7.6 Mobile IP

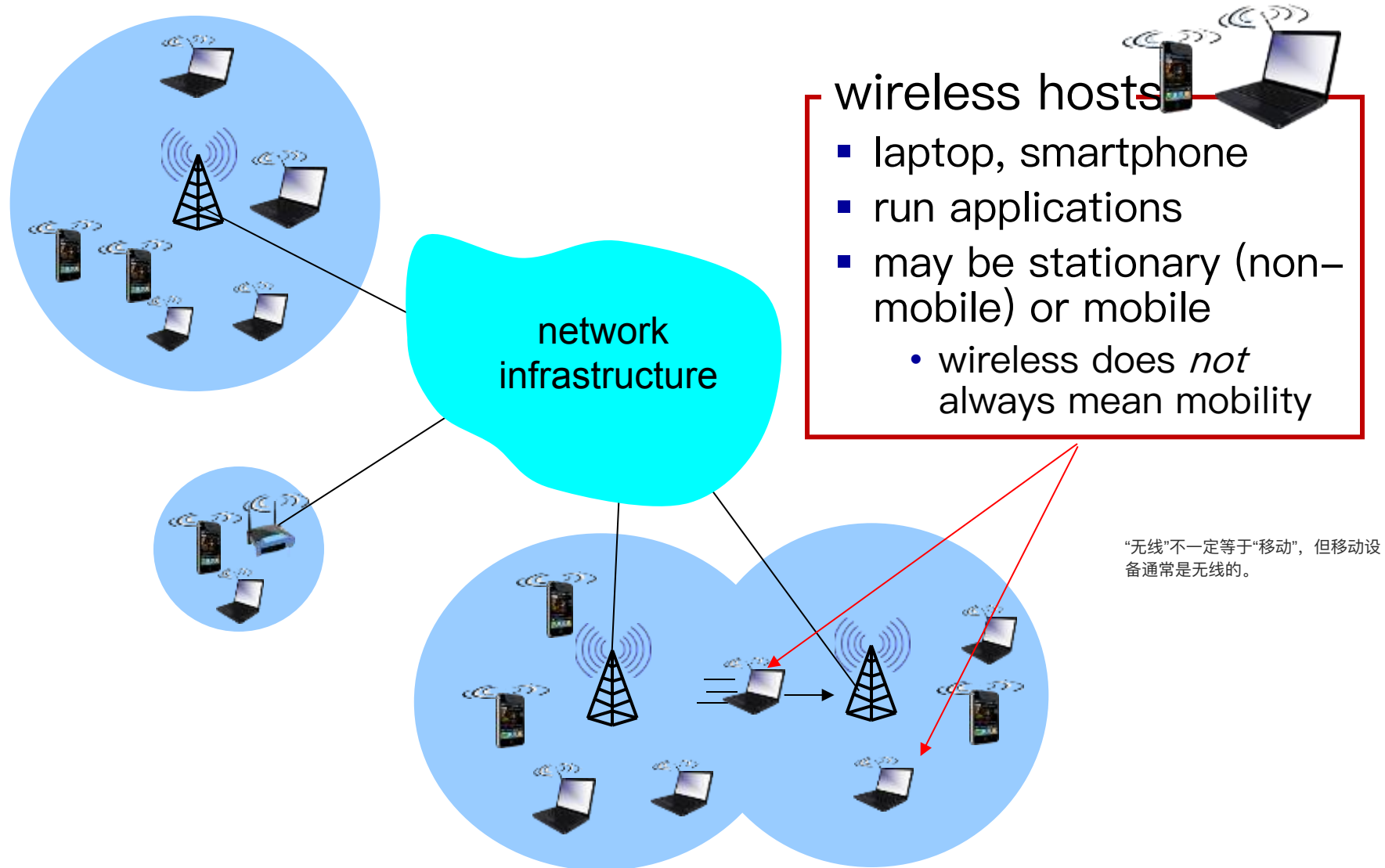
7.7 Handling mobility in cellular networks

7.8 Mobility and higher-layer protocols

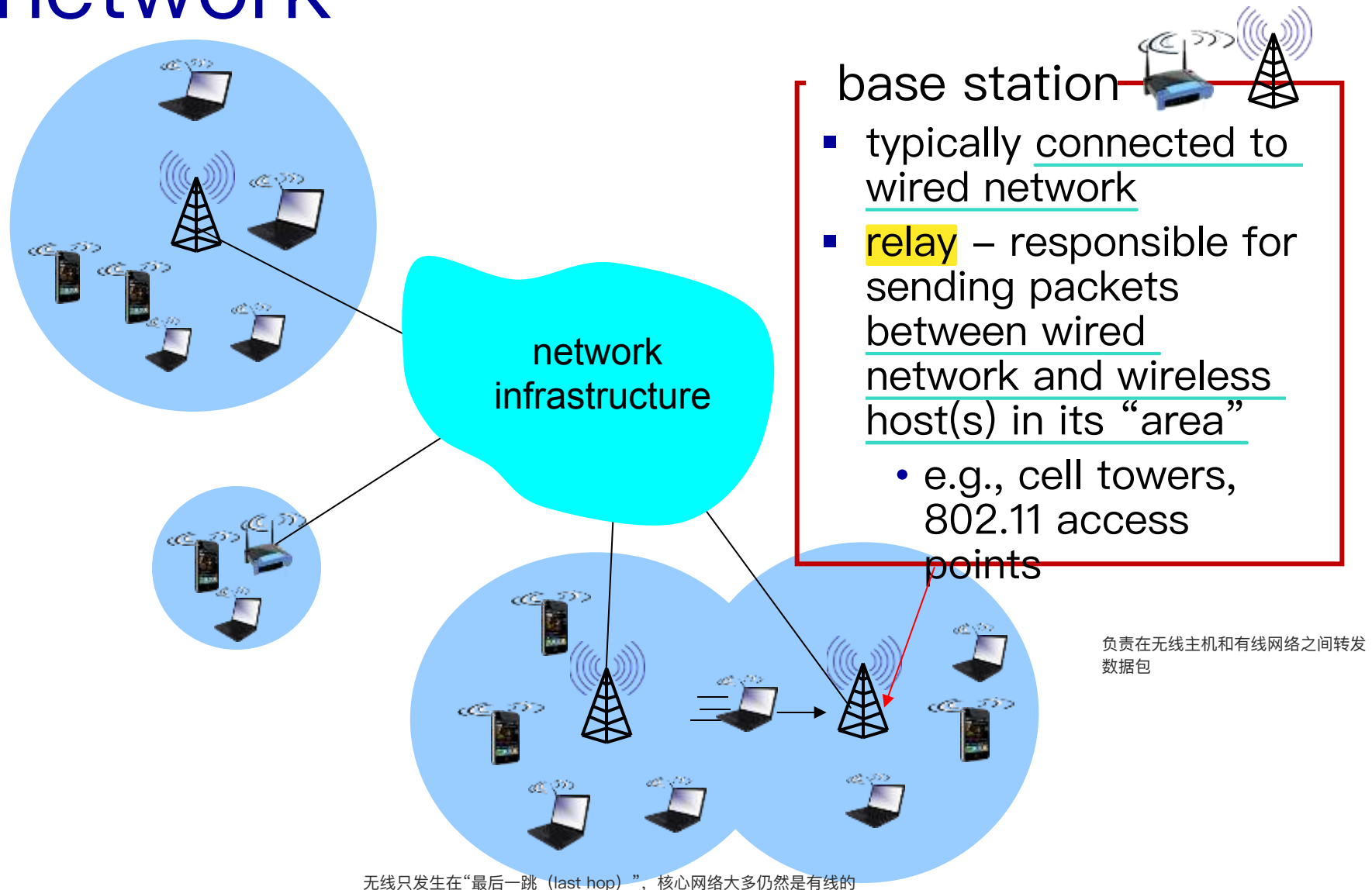
Elements of a wireless network



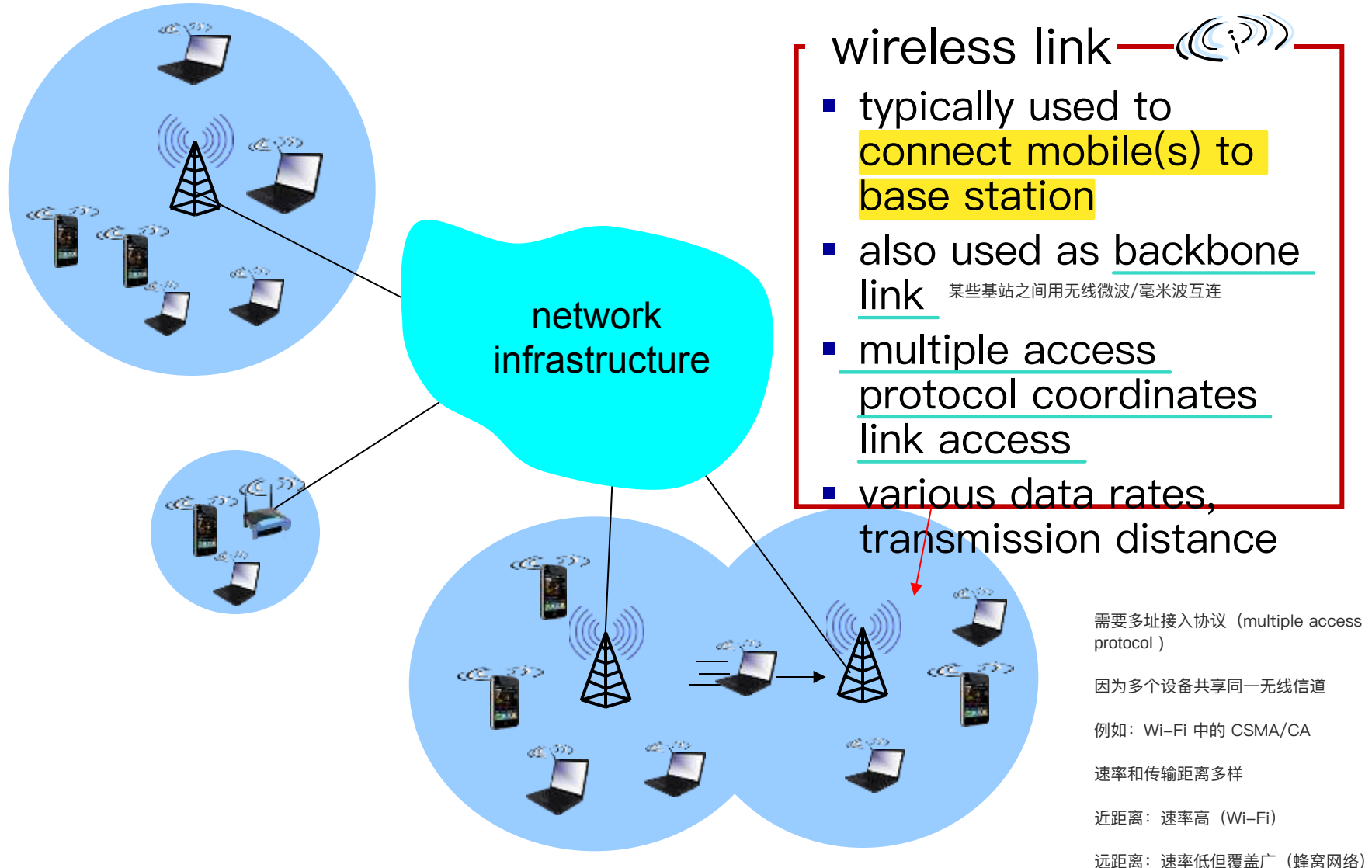
Elements of a wireless network



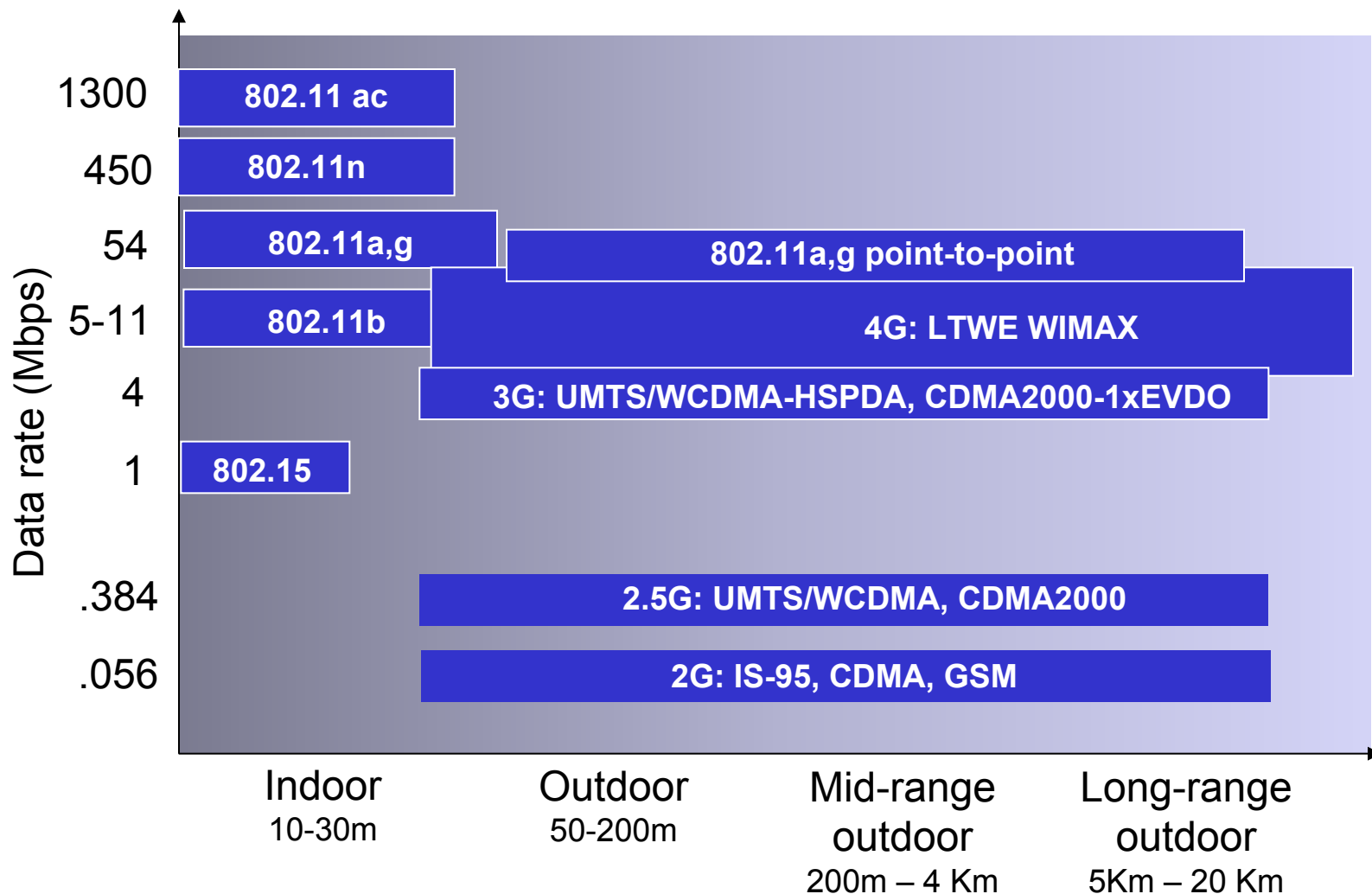
Elements of a wireless network



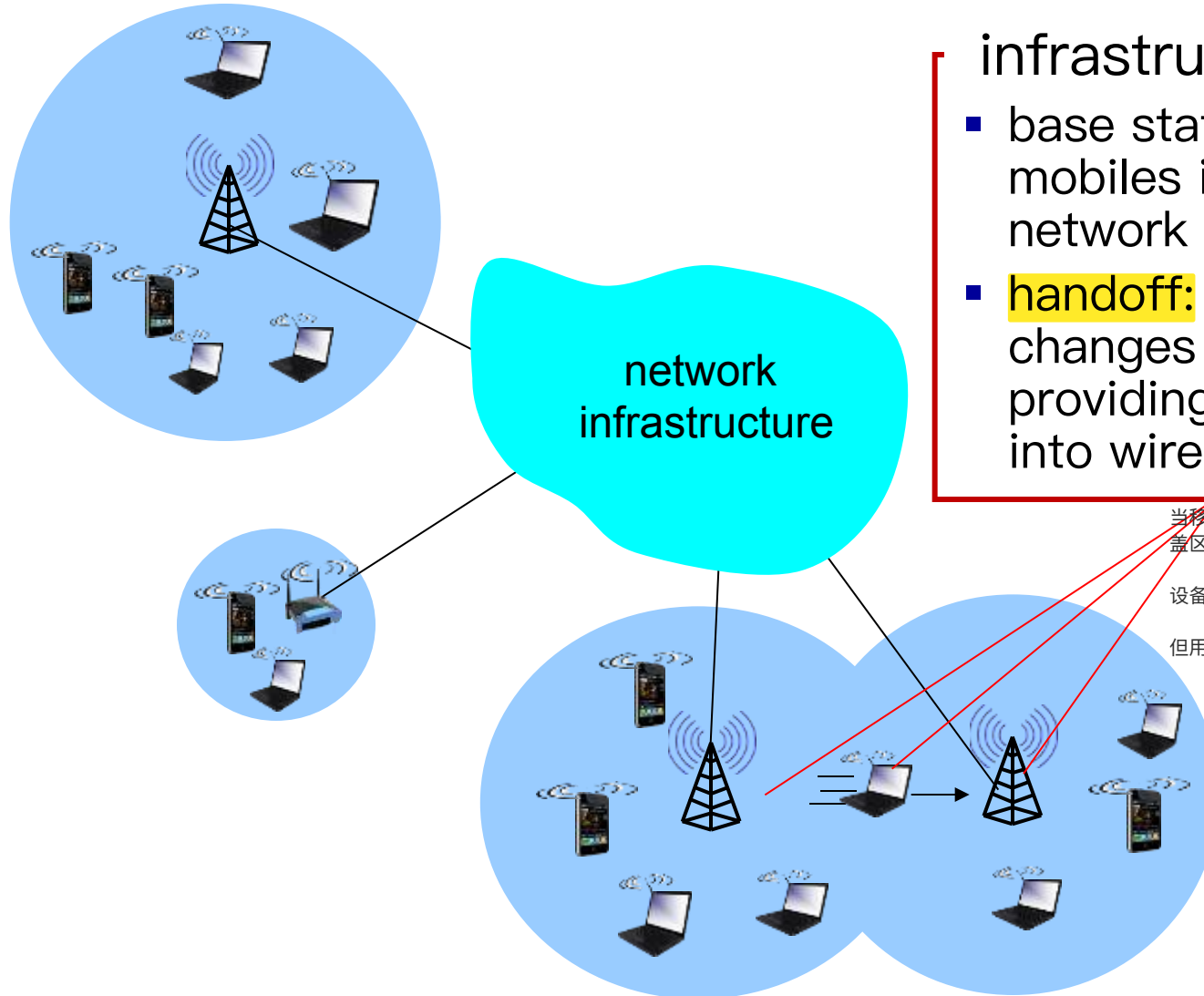
Elements of a wireless network



Characteristics of selected wireless links



Elements of a wireless network



infrastructure mode

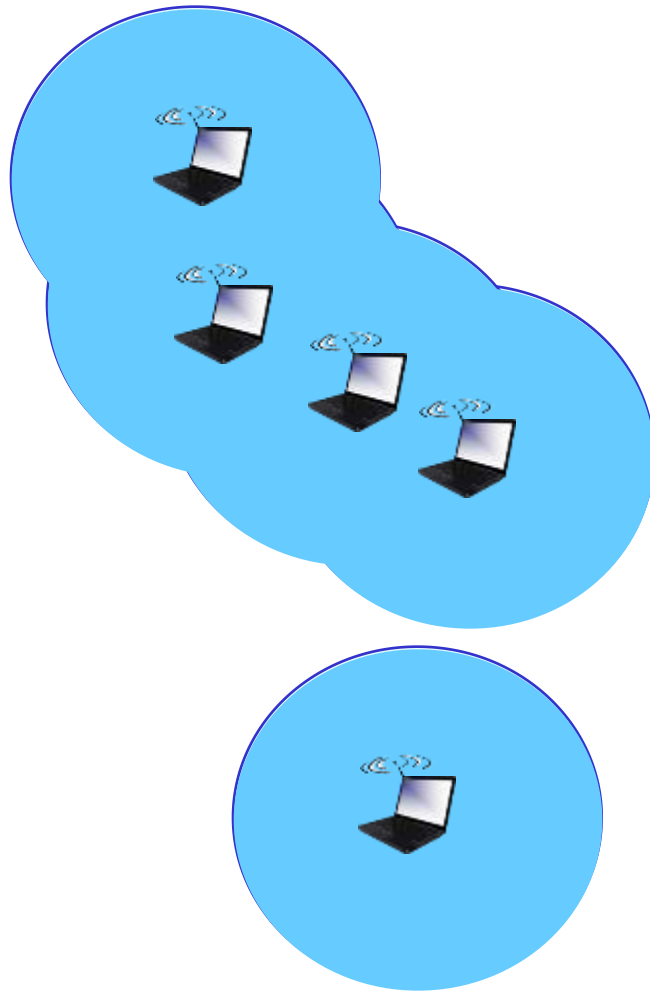
- base station connects mobiles into wired network
- **handoff**: mobile changes base station providing connection into wired network

当移动设备从一个基站的覆盖区走到另一个基站的覆盖区时:

设备会更换提供连接的基站

但用户希望: 连接不中断

Elements of a wireless network



节点只能和:

在自己无线覆盖范围内的节点直接通信

如果目标节点不在范围内:

通过其他节点中继转发 (多跳)

自组织模式

ad hoc mode

- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: route among themselves

节点既是:

终端 (host)

又是路由器 (router)

Wireless network taxonomy

	single hop	multiple hops
infrastructure (e.g., APs)	host connects to base station (WiFi, WiMAX, cellular) <u>which connects to</u> <u>larger Internet</u>	host may <u>have to</u> <u>relay through several</u> <u>wireless nodes to</u> connect to larger Internet: <i>mesh net</i>
no infrastructure	no base station, no connection to larger Internet (Bluetooth, ad hoc nets)	no base station, no connection to larger Internet. May have to relay to reach other a given wireless node <u>MANET, VANET</u>

Chapter 7 outline

7.1 Introduction

Wireless

7.2 Wireless links, characteristics

- CDMA

7.3 IEEE 802.11 wireless LANs (“Wi-Fi”)

7.4 Cellular Internet Access

- architecture
- standards (e.g., 3G, LTE)

Mobility

7.5 Principles: addressing and routing to mobile users

7.6 Mobile IP

7.7 Handling mobility in cellular networks

7.8 Mobility and higher-layer protocols

Wireless Link Characteristics (1)

important differences from wired link

- *decreased signal strength*: radio signal attenuates as it propagates through matter (path loss)
- *interference from other sources*: standardized wireless network frequencies (e.g., 2.4 GHz) shared by other devices (e.g., phone); devices (motors) interfere as well
- *multipath propagation*: radio signal reflects off objects ground, arriving at destination at slightly different times

多条路径 → 到达时间略有不同

造成 时变信道

.... make communication across (even a point to point) wireless link much more “difficult”

Wireless Link Characteristics (2)

- SNR: signal-to-noise ratio
 - larger SNR – easier to extract signal from noise (a “good thing”)

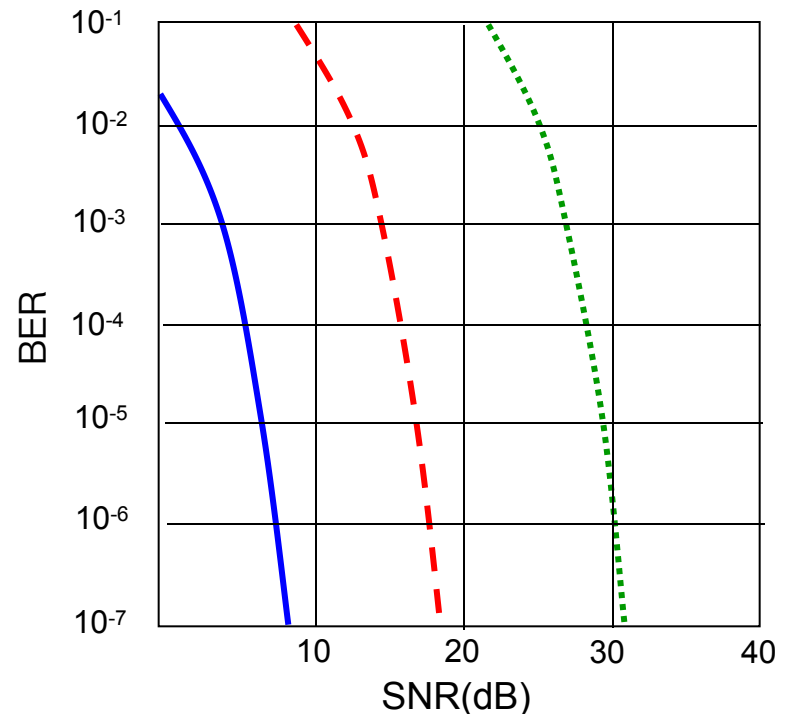
BER : 误码率 (Bit Error Rate)

- *SNR versus BER tradeoffs*

- *given physical layer*: increase power $\rightarrow$ increase SNR $\rightarrow$ decrease BER
- *given SNR*: choose physical layer that meets BER requirement, giving highest throughput
 - SNR may change with mobility: dynamically adapt physical layer (modulation technique, rate)

动态调整调制方式和速率

Wi-Fi / 4G / 5G 都在做这件事



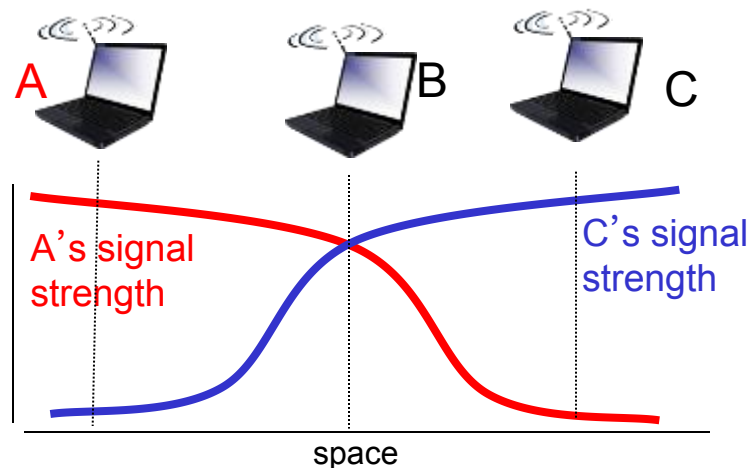
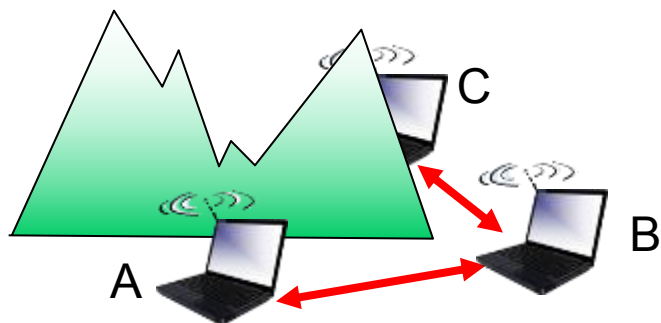
..... QAM256 (8 Mbps)

- - - QAM16 (4 Mbps)

— BPSK (1 Mbps)

Wireless network characteristics

Multiple wireless senders and receivers create additional problems (beyond multiple access):



Hidden terminal problem

- B, A hear each other
- B, C hear each other
- A, C can not hear each other means A, C unaware of their interference at B

A 和 C:

都不知道自己干扰了对方

因为它们“互相听不见”

这就是 CSMA 在无线中失效的原因

Fading:

- B, A hear each other
- B, C hear each other
- A, C can not hear each other interfering at B

Code Division Multiple Access (CDMA)

- unique “code” assigned to each user; i.e., **code set partitioning**

- all users share same frequency, but **each user has own “chipping” sequence** (i.e., code) to encode data
- allows multiple users to “coexist” and transmit simultaneously with minimal interference (if codes are **“orthogonal”**)

所有用户共享同一频率、同一时间在发信号
但每个人用不同的“码序列”把数据“扩频”后再发送

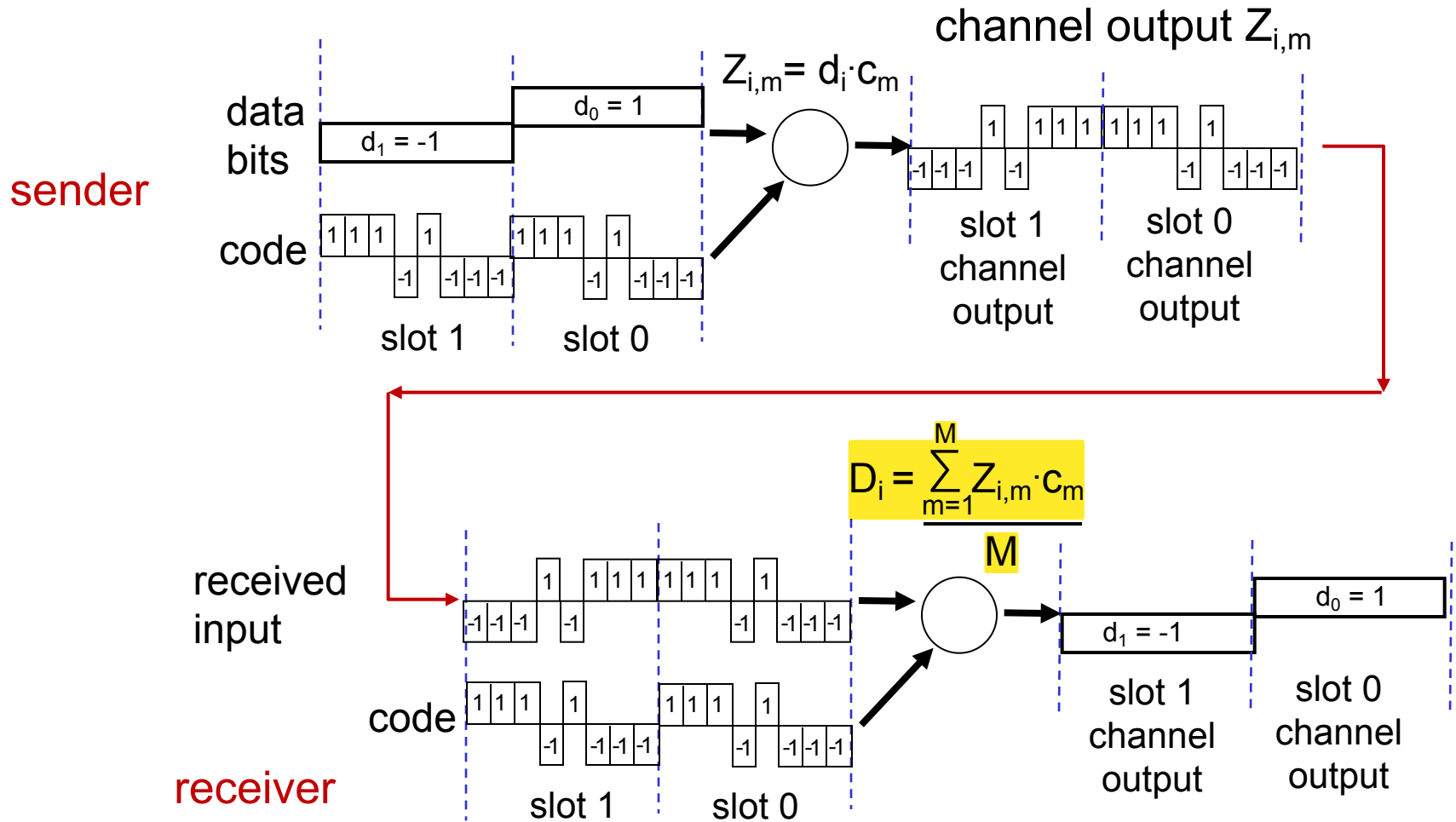
如果这些码彼此正交/相关性很低，就能“同时共存”且干扰小

- *encoded signal* = (original data) X (chipping sequence)

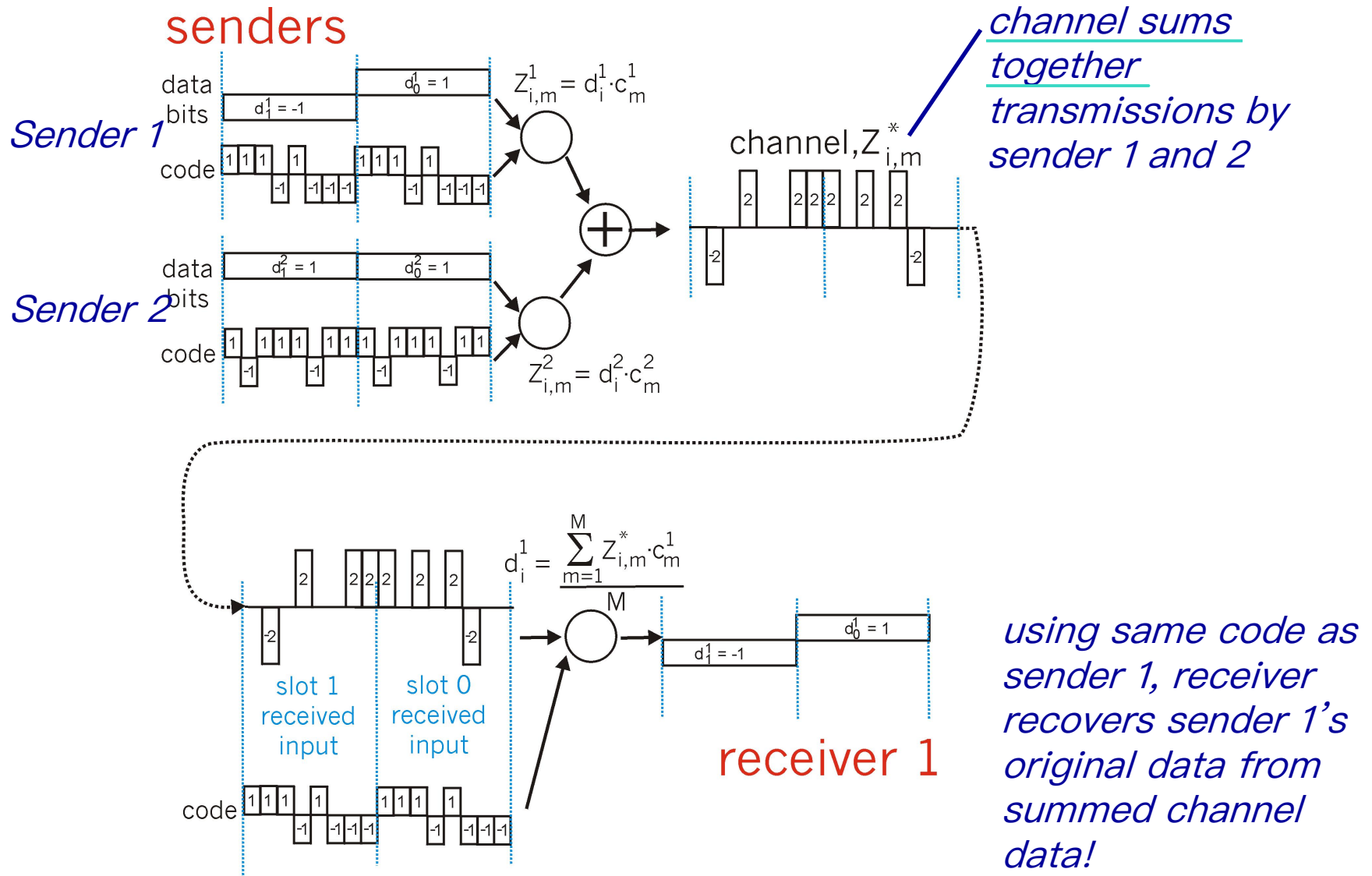
逐chip相乘

- *decoding*: **inner-product** of encoded signal and chipping sequence

CDMA encode/decode



CDMA: two-sender interference



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IEEE 802.11 Wireless LAN

802.11b

- 2.4–5 GHz unlicensed spectrum
- up to 11 Mbps
- direct sequence spread spectrum (DSSS) in physical layer
 - all hosts use same chipping code

802.11a

- 5–6 GHz range
- up to 54 Mbps

802.11g

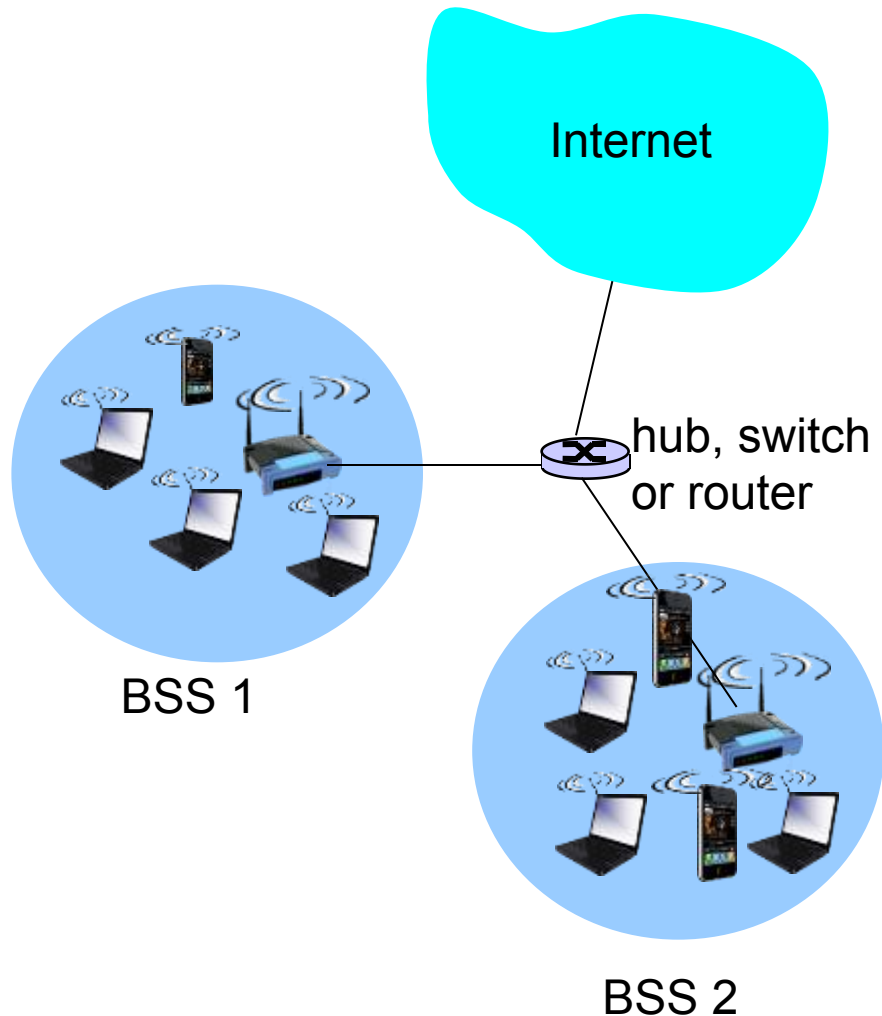
- 2.4–5 GHz range
- up to 54 Mbps

802.11n: multiple antennae

- 2.4–5 GHz range
- up to 200 Mbps

-
- all use CSMA/CA for multiple access
 - all have base-station and ad-hoc network versions

802.11 LAN architecture

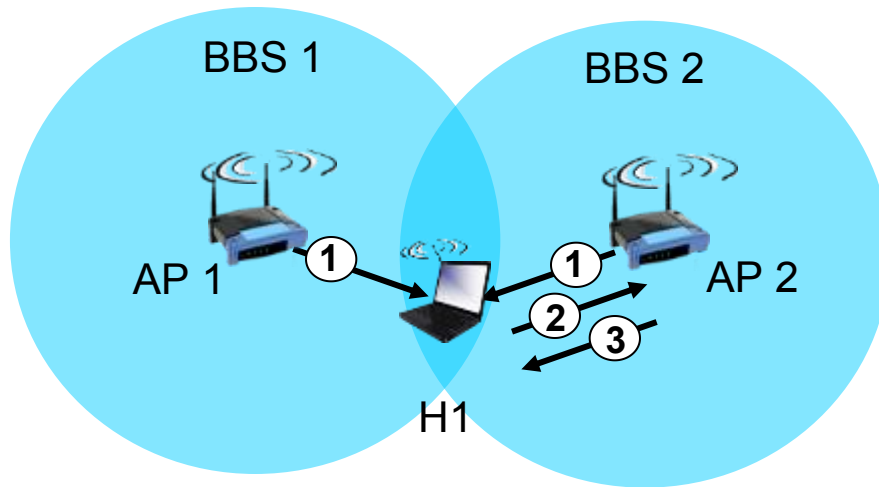


- wireless host communicates with base station
 - base station = access point (AP)
- **Basic Service Set (BSS)** (aka “cell”) in infrastructure mode contains:
 - wireless hosts
 - access point (AP): base stationad hoc mode: hosts only

802.11: Channels, association

- 802.11b: 2.4GHz–2.485GHz spectrum divided into 11 channels at different frequencies
 - AP admin chooses frequency for AP
 - 邻居家 AP 可能:
选了同一个信道
 - 或选了部分重叠的信道
 - interference possible: channel can be same as that chosen by neighboring AP!
- host: **must *associate* with an AP**
 - scans channels, listening for *beacon frames*
containing AP's name (SSID) and MAC address
 - selects AP to associate with
 - may perform authentication [Chapter 8]
 - will typically run DHCP to get IP address in AP's subnet

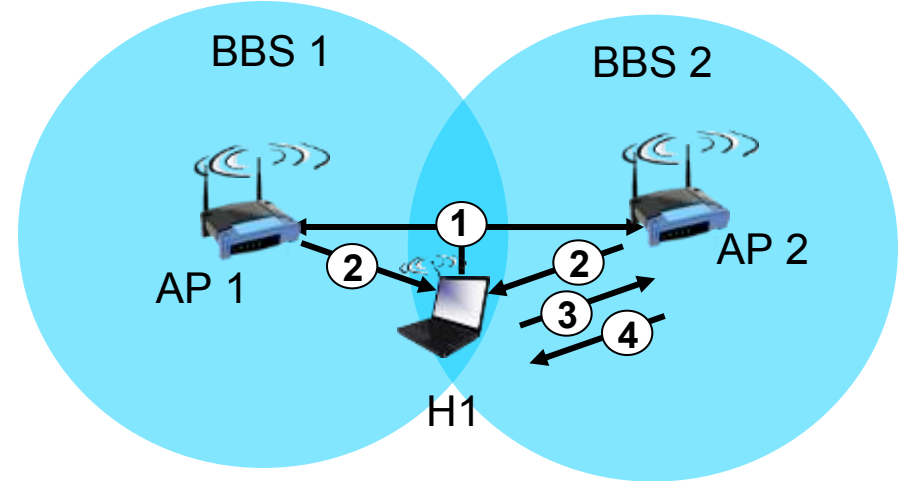
802.11: passive/active scanning



passive scanning:

- (1) beacon frames sent from APs
- (2) association Request frame sent:
H1 to selected AP
- (3) association Response frame sent
from selected AP to H1

慢、省能



active scanning:

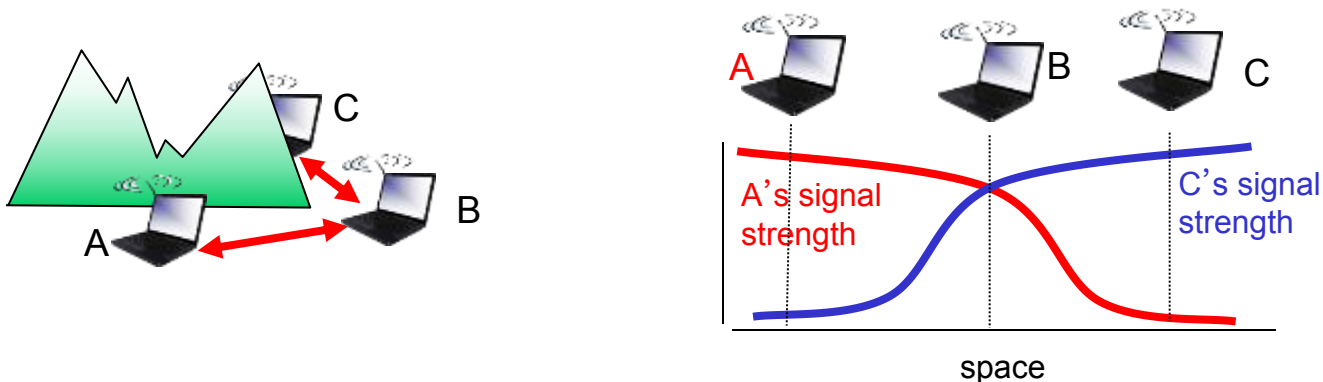
- (1) **Probe Request frame broadcast**
from H1
- (2) Probe Response frames sent
from APs
- (3) Association Request frame sent:
H1 to selected AP
- (4) Association Response frame sent
from selected AP to H1

快、费能

IEEE 802.11: multiple access

- avoid collisions: 2+ nodes transmitting at same time
- 802.11: CSMA – **sense before transmitting**
 - don't collide with ongoing transmission by other node
- 802.11: no collision detection!
 - difficult to receive (sense collisions) when transmitting due to weak received signals (fading)
 - can't sense all collisions in any case: hidden terminal, fading
 - goal: **avoid collisions**: CSMA/C(ollision)A(avoidance)

以太网是 CSMA/CD
Wi-Fi 是 CSMA/CA



IEEE 802.11 MAC Protocol: CSMA/CA

802.11 sender

- 1 if sense channel idle for **DIFS** then
transmit entire frame (no CD)
- 2 if sense channel busy then
start random backoff time
timer counts down while channel idle
transmit when timer expires
if no ACK, increase random backoff interval,
repeat 2

Distributed Inter-Frame Space

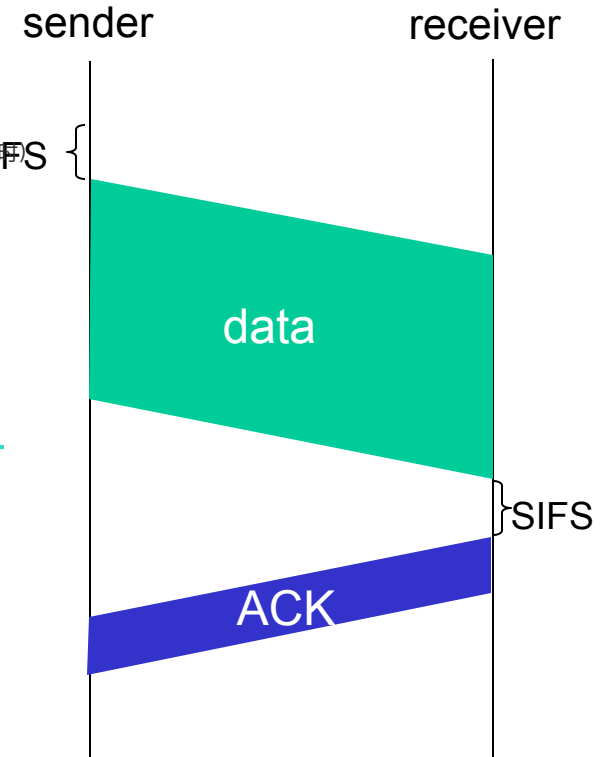
信道必须连续空闲一个 DIFS 时

开始倒计时 (只在信道空闲时)
信道空闲 → 计数器减 1
信道变忙 → 冻结计数器
再次空闲 → 继续倒计时

802.11 receiver

- if frame received OK
return ACK after **SIFS** (ACK needed due to hidden terminal problem)

→ ACK 拥有更高优先级
→ 别人来不及插队



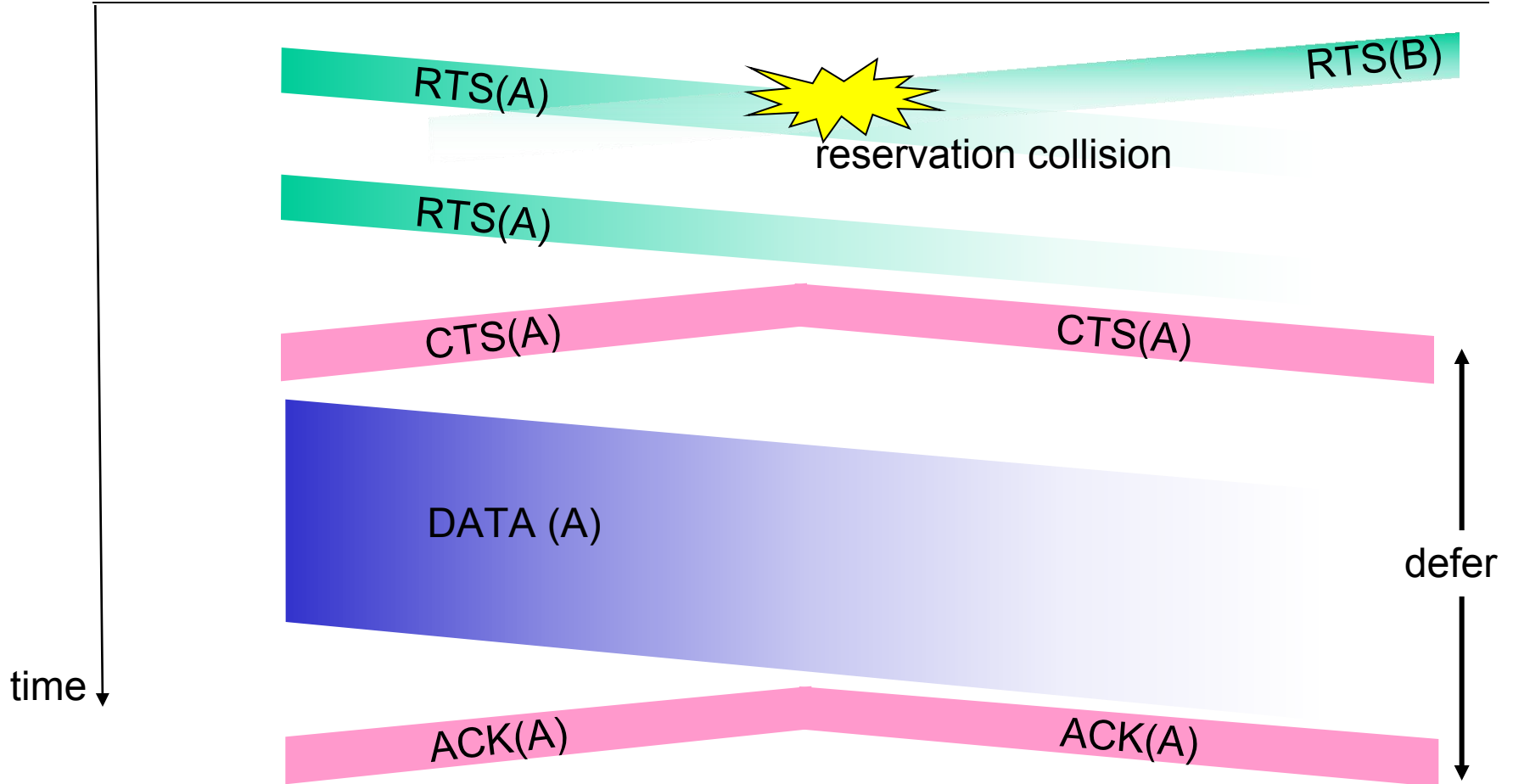
Avoiding collisions (more)

idea: allow sender to “reserve” channel rather than random access of data frames: avoid collisions of long data frames

- sender first transmits small request-to-send (RTS) packets to BS using CSMA
 - RTSs may still collide with each other (but they’re short)
- BS broadcasts clear-to-send CTS in response to RTS
- **CTS heard by all nodes**
 - sender transmits data frame
 - other stations defer transmissions

*avoid data frame collisions completely
using small reservation packets!*

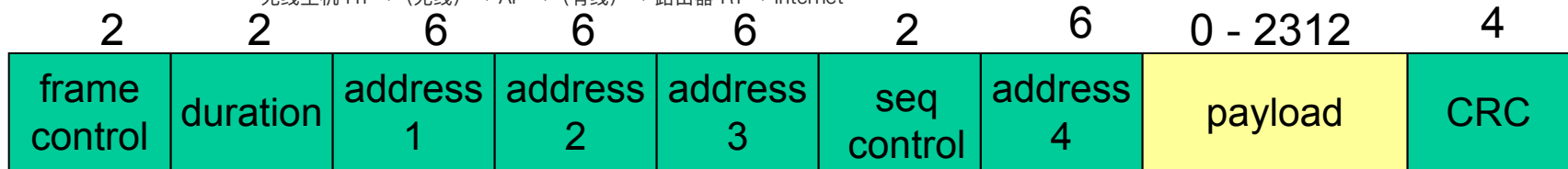
Collision Avoidance: RTS-CTS ~~exchange~~



802.11 frame: addressing

但 Wi-Fi 的数据经常是这种路径：

无线主机 H1 → (无线) → AP → (有线) → 路由器 R1 → Internet



Address 1: MAC address of wireless host or AP to receive this frame

Address 2: MAC address of wireless host or AP transmitting this frame

Address 3: MAC address of router interface to which AP is attached

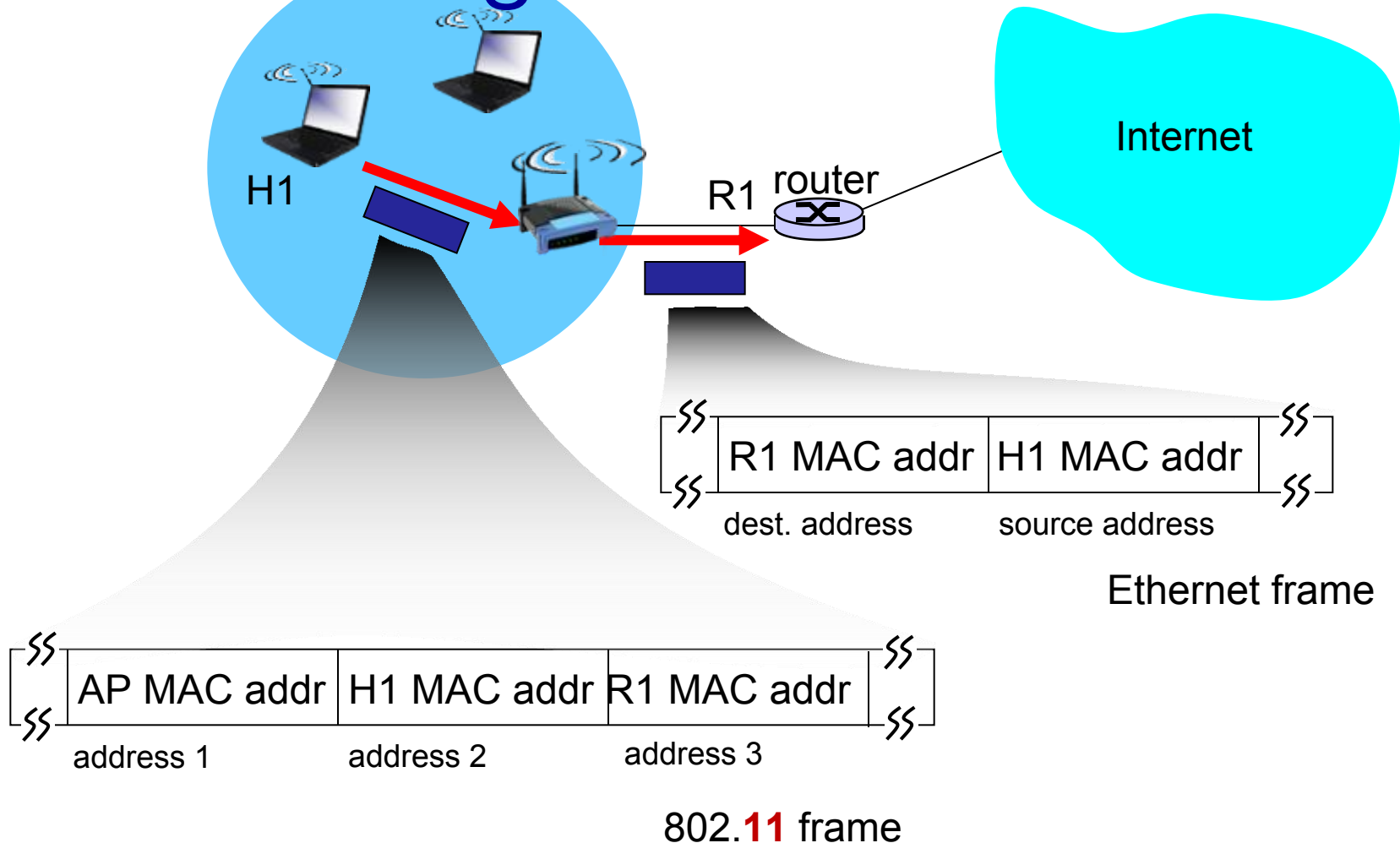
Address 4: used only in ad hoc mode

注意：

在无线这段里，真正“接收你无线电波”的是 AP；

但从“网络转发”的角度，这个帧最终是要交给路由器 R1（或某个有线侧目的主机）。

802.11 frame: addressing



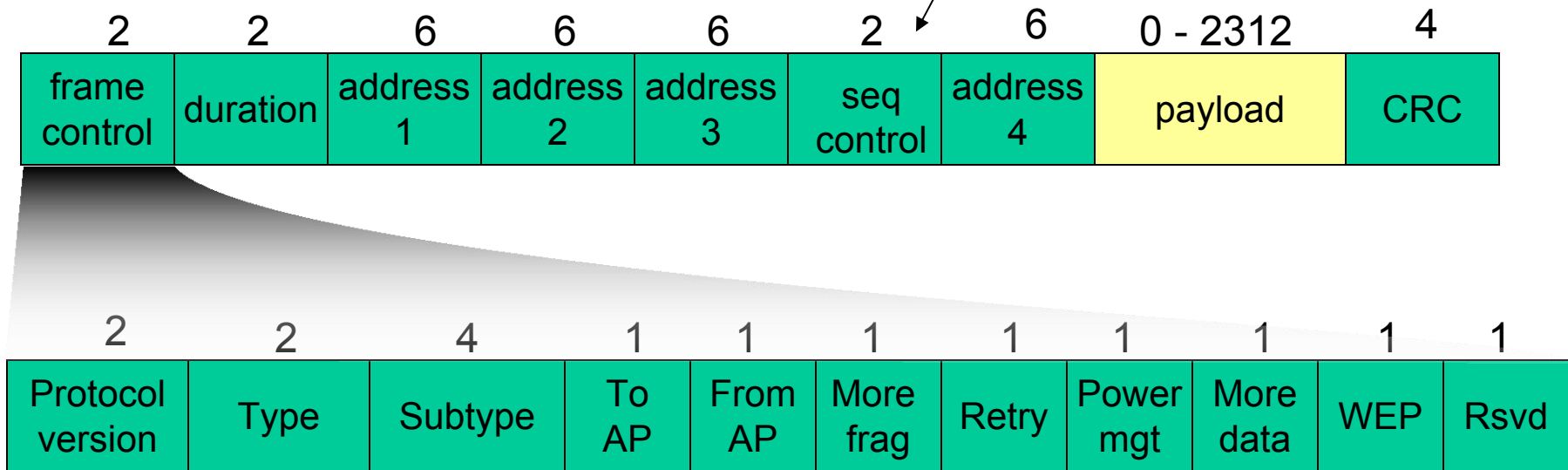
802.11 frame: more

信道接下来还要被占用多久，你们先别抢。

duration of reserved
transmission time (RTS/CTS)

frame seq #
(for RDT)

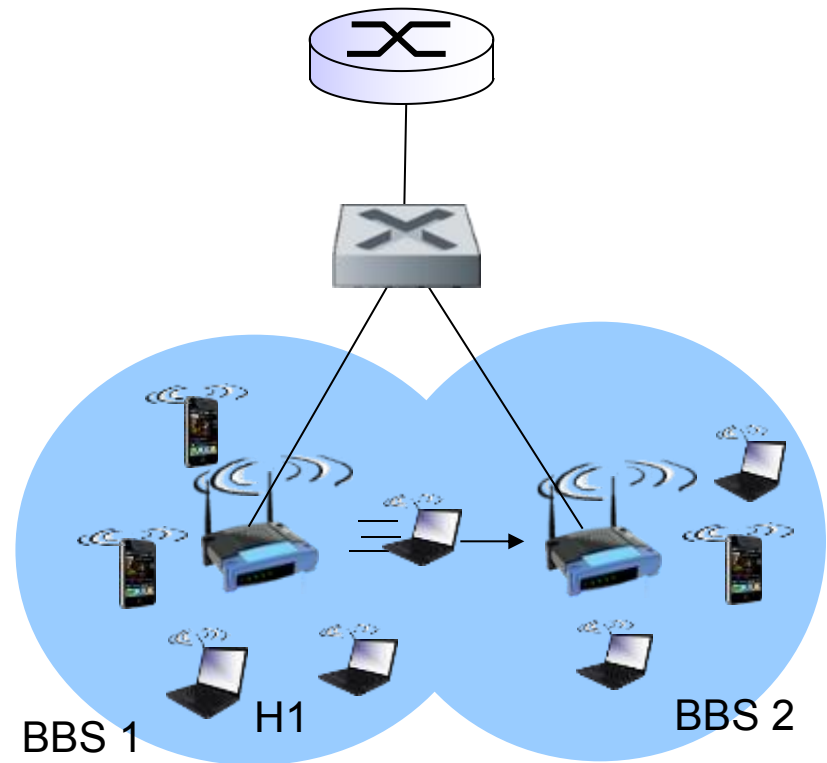
防止重复帧（因为没收到 ACK 会重传）



frame type
(RTS, CTS, ACK, data)

802.11: mobility within same subnet

- H1 remains in same IP subnet: IP address can remain same
- switch: which AP is associated with H1?
 - self-learning (Ch. 5): switch will see frame from H1 and “remember” which switch port can be used to reach H1

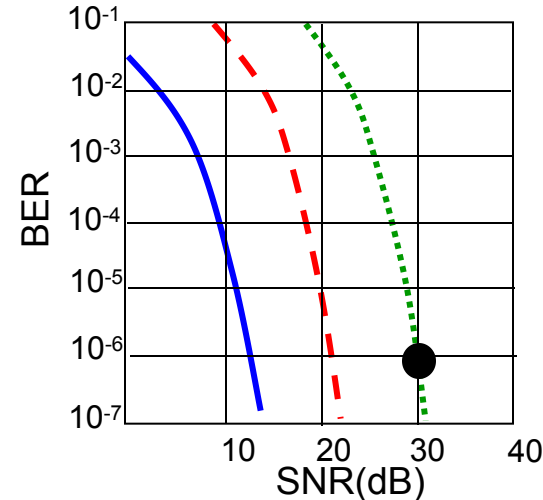
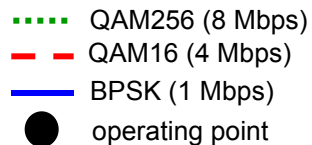


802.11: advanced capabilities

人/设备移动 → 无线信道变差 (SNR 下降) → 错误率 BER 上升 → 需要降速率来保证可靠性

Rate adaptation

- base station, mobile dynamically change transmission rate (physical layer modulation technique) as mobile moves, SNR varies



- SNR decreases, BER increase as node moves away from base station
- When BER becomes too high, switch to lower transmission rate but with lower BER

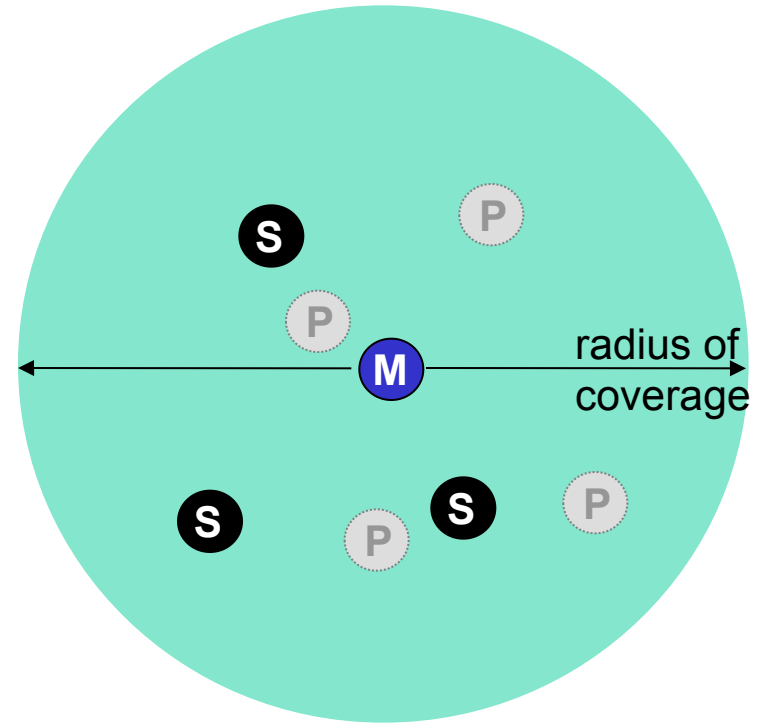
802.11: advanced capabilities

power management

- node-to-AP: “I am going to sleep until next beacon frame”
 - AP knows not to transmit frames to this node
AP 不会直接把数据发给它，而是先缓存下来。
 - node wakes up before next beacon frame
- beacon frame: contains list of mobiles with AP-to-mobile frames waiting to be sent
哪些移动设备（手机/笔记本）在 AP 那里有下行数据帧（AP→mobile）已经缓存着、等着发
 - node will stay awake if AP-to-mobile frames to be sent; otherwise sleep again until next beacon frame

802.15: personal area network

- less than 10 m diameter
- replacement for cables (mouse, keyboard, headphones)
- ad hoc: no infrastructure
- master/slaves:
 - slaves request permission to send (to master)
 - master grants requests
- 802.15: evolved from Bluetooth specification
 - 2.4–2.5 GHz radio band
 - up to 721 kbps



- (M)** Master device
- (S)** Slave device
- (P)** Parked device (inactive)